

```
In [ ]: pip install pandas seaborn matplotlib
```

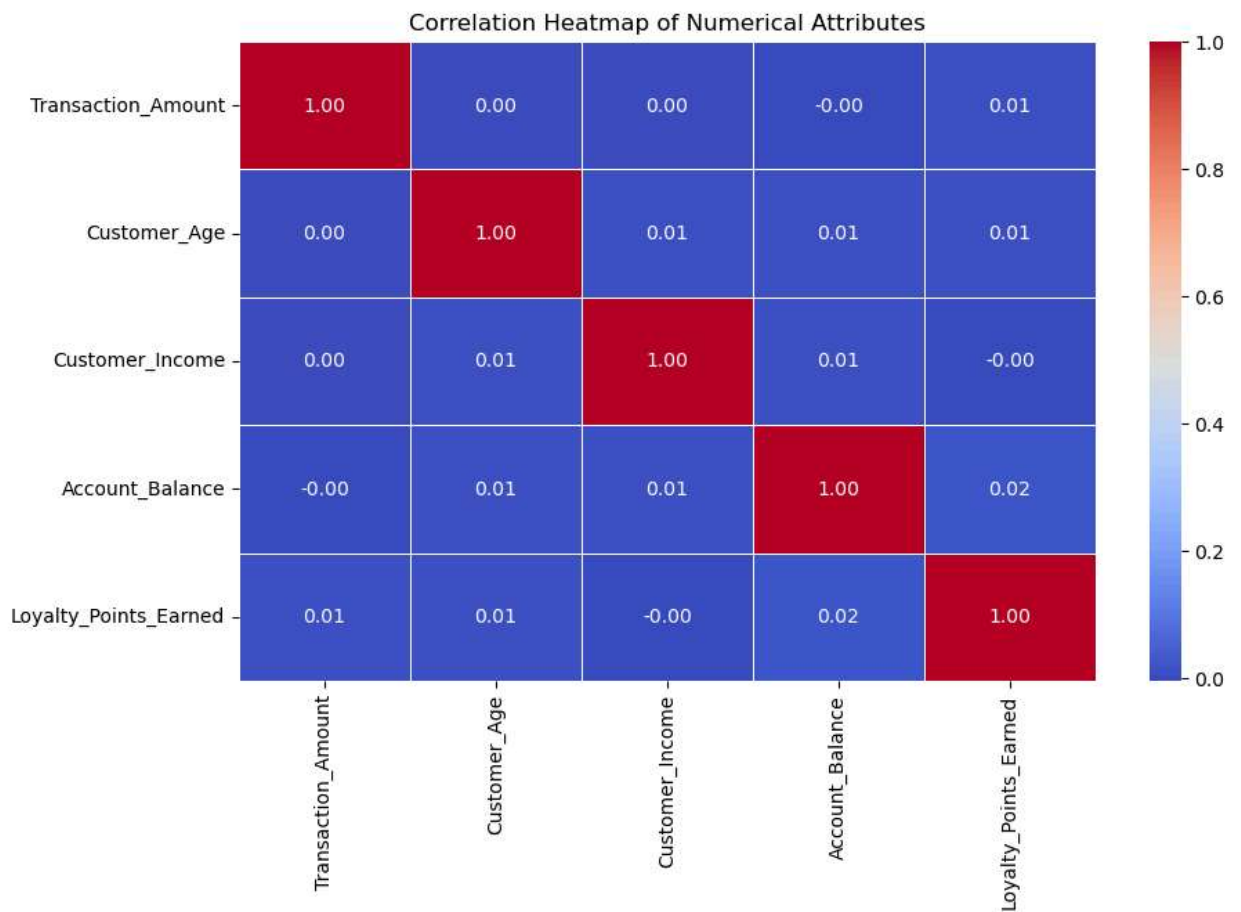
```
In [3]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load the dataset
file_path = r"C:\Users\hethv\OneDrive\Desktop\Banking_Transactions_USA_2023_2024.csv"
# Update with your correct path
df = pd.read_csv(file_path)

# Selecting only numerical attributes
numerical_df = df.select_dtypes(include=['float64', 'int64'])

# Compute correlation matrix
correlation_matrix = numerical_df.corr()

# Plot heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap of Numerical Attributes")
plt.show()
```



```
In [4]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
```

```

# Load the dataset
file_path = r"C:\Users\hethv\OneDrive\Desktop\Banking_Transactions_USA_2023_2024.csv"
df = pd.read_csv(file_path)

# Selecting only categorical attributes
categorical_cols = df.select_dtypes(include=['object', 'bool']).columns

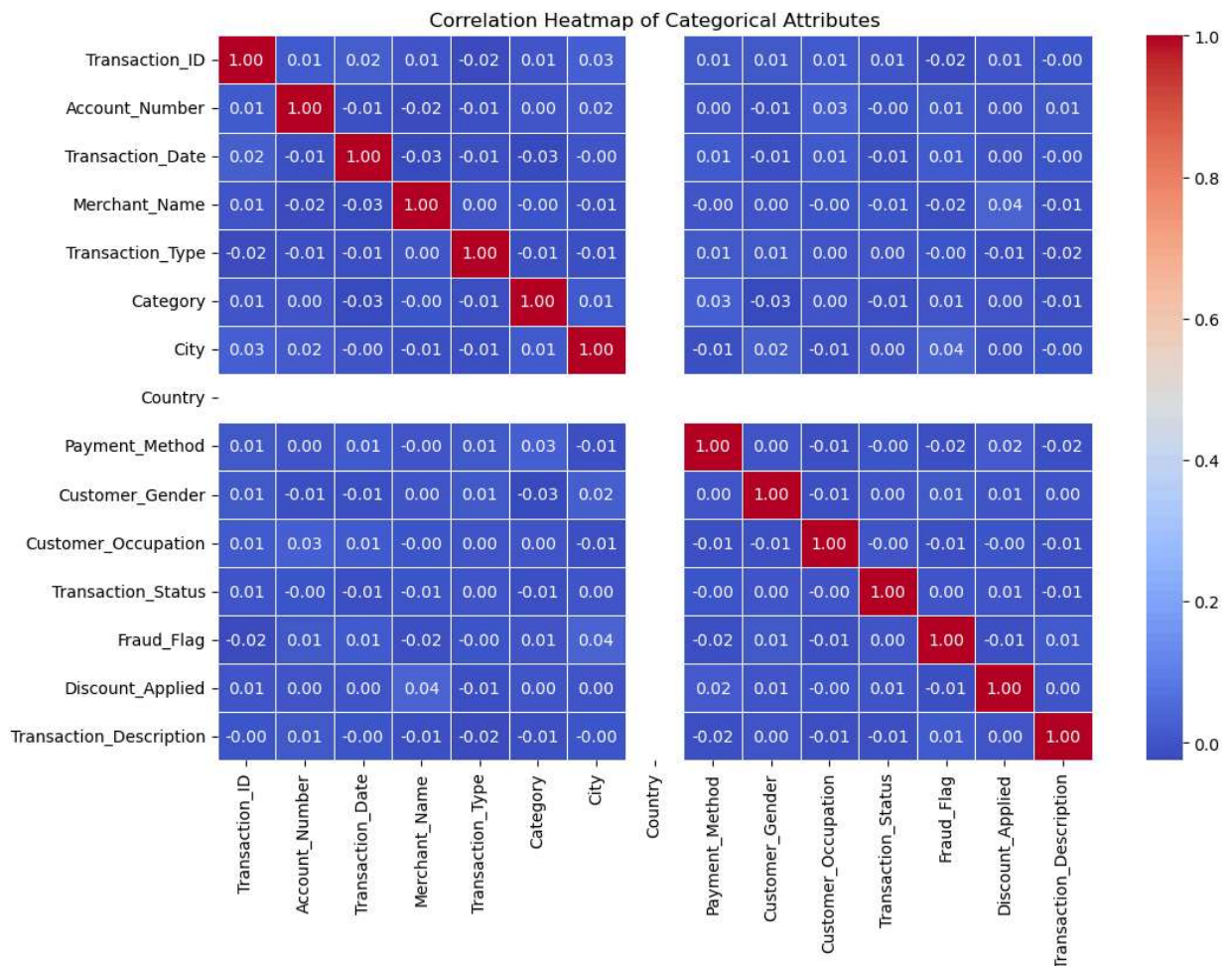
# Convert categorical attributes to numerical using Label Encoding
label_encoders = {}
encoded_df = df.copy()

for col in categorical_cols:
    le = LabelEncoder()
    encoded_df[col] = le.fit_transform(df[col])
    label_encoders[col] = le # Store encoders if needed later

# Compute correlation matrix
correlation_matrix = encoded_df[categorical_cols].corr()

# Plot heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap of Categorical Attributes")
plt.show()

```



```

In [5]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder

```

```

# Load the dataset
file_path = r"C:\Users\hethv\OneDrive\Desktop\Banking_Transactions_USA_2023_2024.csv"
df = pd.read_csv(file_path)

# Identify categorical and numerical attributes
categorical_cols = df.select_dtypes(include=['object', 'bool']).columns
numerical_cols = df.select_dtypes(include=['float64', 'int64']).columns

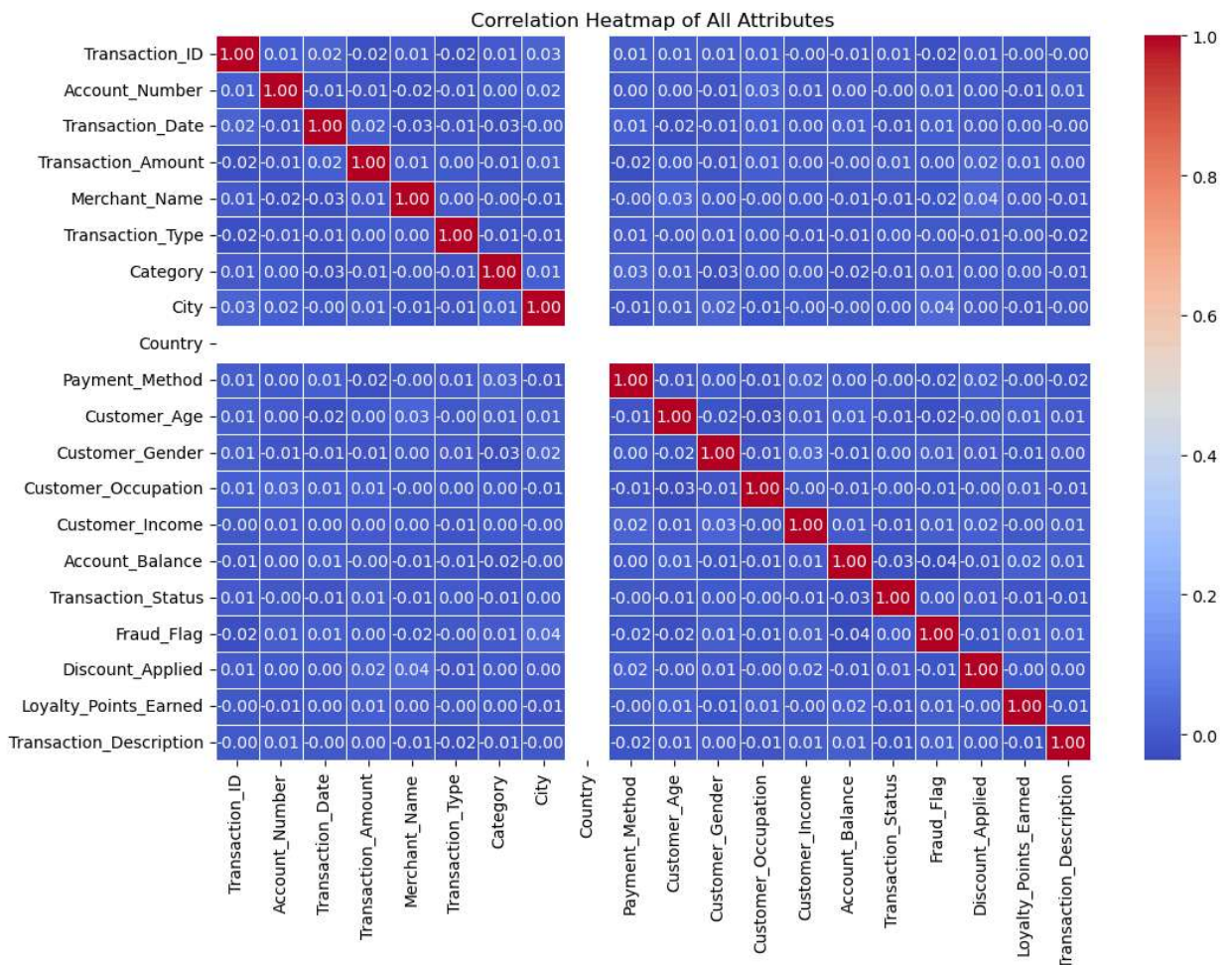
# Convert categorical attributes to numerical using Label Encoding
label_encoders = {}
encoded_df = df.copy()

for col in categorical_cols:
    le = LabelEncoder()
    encoded_df[col] = le.fit_transform(df[col])
    label_encoders[col] = le # Store encoders if needed later

# Compute correlation matrix for all attributes
correlation_matrix = encoded_df.corr()

# Plot heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap of All Attributes")
plt.show()

```



```

In [6]: import pandas as pd
import seaborn as sns

```

```
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder

# Load the dataset
file_path = r"C:\Users\hethv\OneDrive\Desktop\Banking_Transactions_USA_2023_2024.csv"
df = pd.read_csv(file_path)

# Reduce the dataset by taking a random sample (e.g., 1000 rows)
df = df.sample(n=1000, random_state=42) # Adjust the sample size as needed

# Identify categorical and numerical attributes
categorical_cols = df.select_dtypes(include=['object', 'bool']).columns
numerical_cols = df.select_dtypes(include=['float64', 'int64']).columns

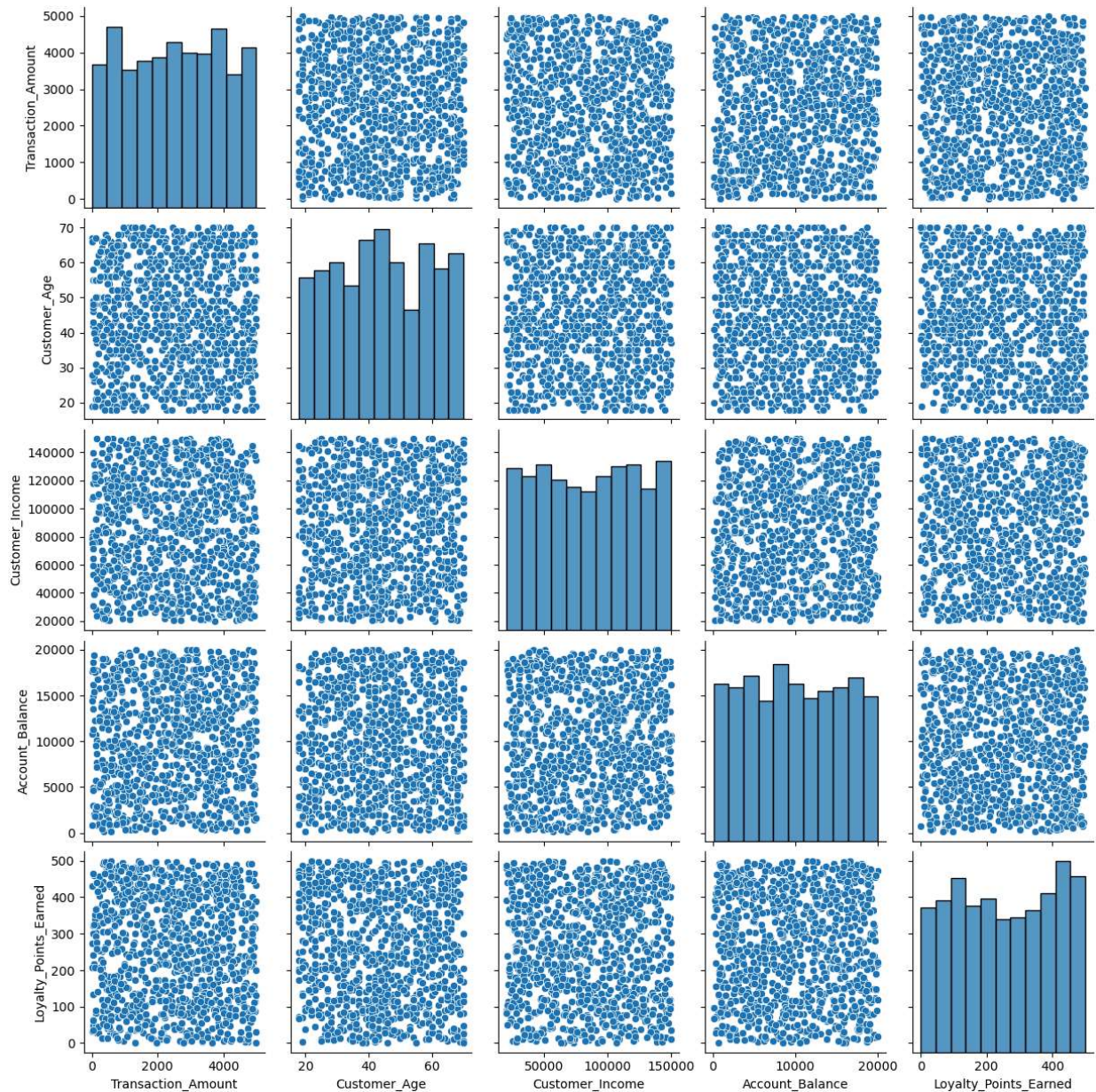
# Convert categorical attributes to numerical using Label Encoding
label_encoders = {}
encoded_df = df.copy()

for col in categorical_cols:
    le = LabelEncoder()
    encoded_df[col] = le.fit_transform(df[col])
    label_encoders[col] = le # Store encoders if needed later

# Reduce the number of attributes to avoid overcrowded plots
selected_cols = numerical_cols[:5] # Select only the first 5 numerical columns for vi

# Scatter plot matrix for selected attributes
sns.pairplot(encoded_df[selected_cols])
plt.show()
```

```
C:\Users\hethv\anaconda3\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The
figure layout has changed to tight
  self._figure.tight_layout(*args, **kwargs)
```

```
In [7]: print(df_sampled.head()) # Displays the first 5 rows
        print(df_sampled.shape) # Shows the number of rows and columns
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[7], line 1
----> 1 print(df_sampled.head()) # Displays the first 5 rows
      2 print(df_sampled.shape)

NameError: name 'df_sampled' is not defined
```

```
In [11]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import os
from sklearn.preprocessing import LabelEncoder

# Step 1: Load the dataset with the correct file path
file_path = r"C:\Users\hethv\OneDrive\Desktop\Banking_Transactions_USA_2023_2024.csv"
```

```

if not os.path.exists(file_path):
    raise FileNotFoundError(f"File not found: {file_path}. Please check the path.")

df = pd.read_csv(file_path)

# Step 2: Convert categorical attributes into numerical using Label Encoding
label_encoders = {}
for col in df.select_dtypes(include=['object', 'bool']).columns:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    label_encoders[col] = le

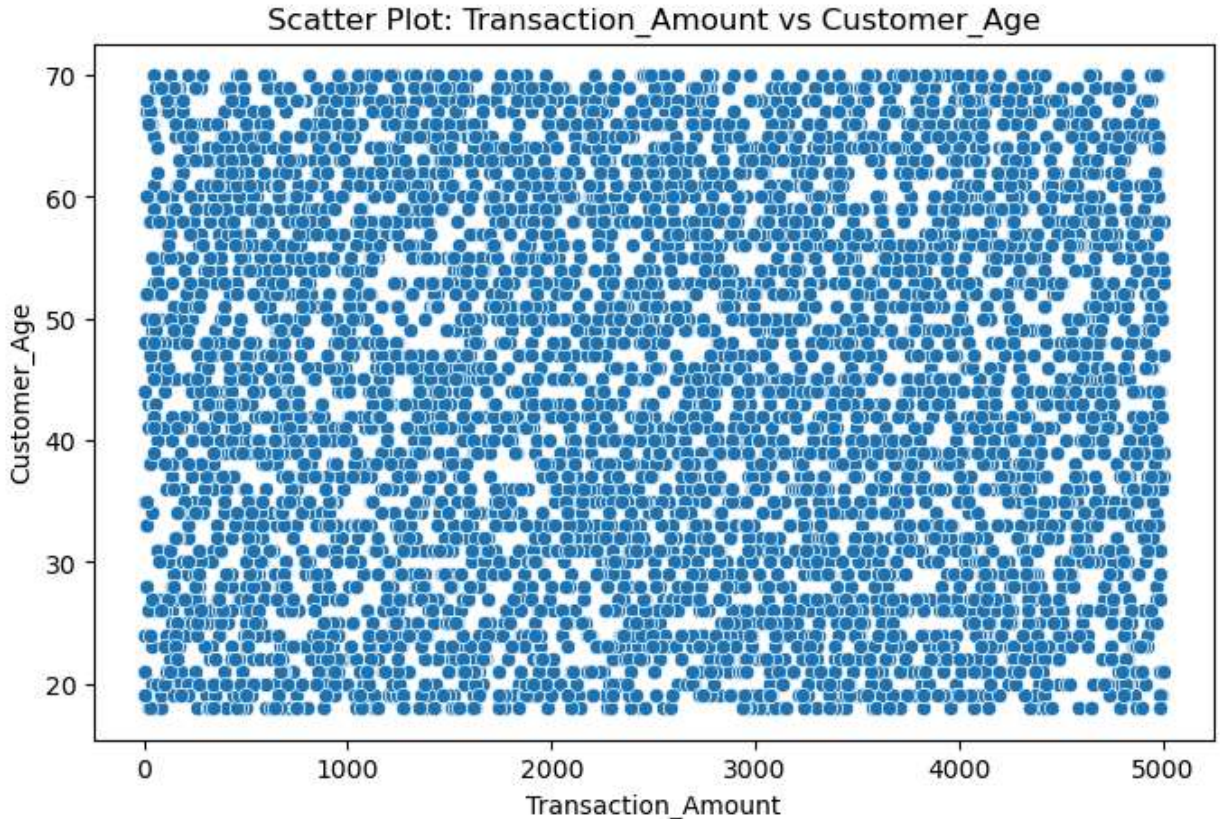
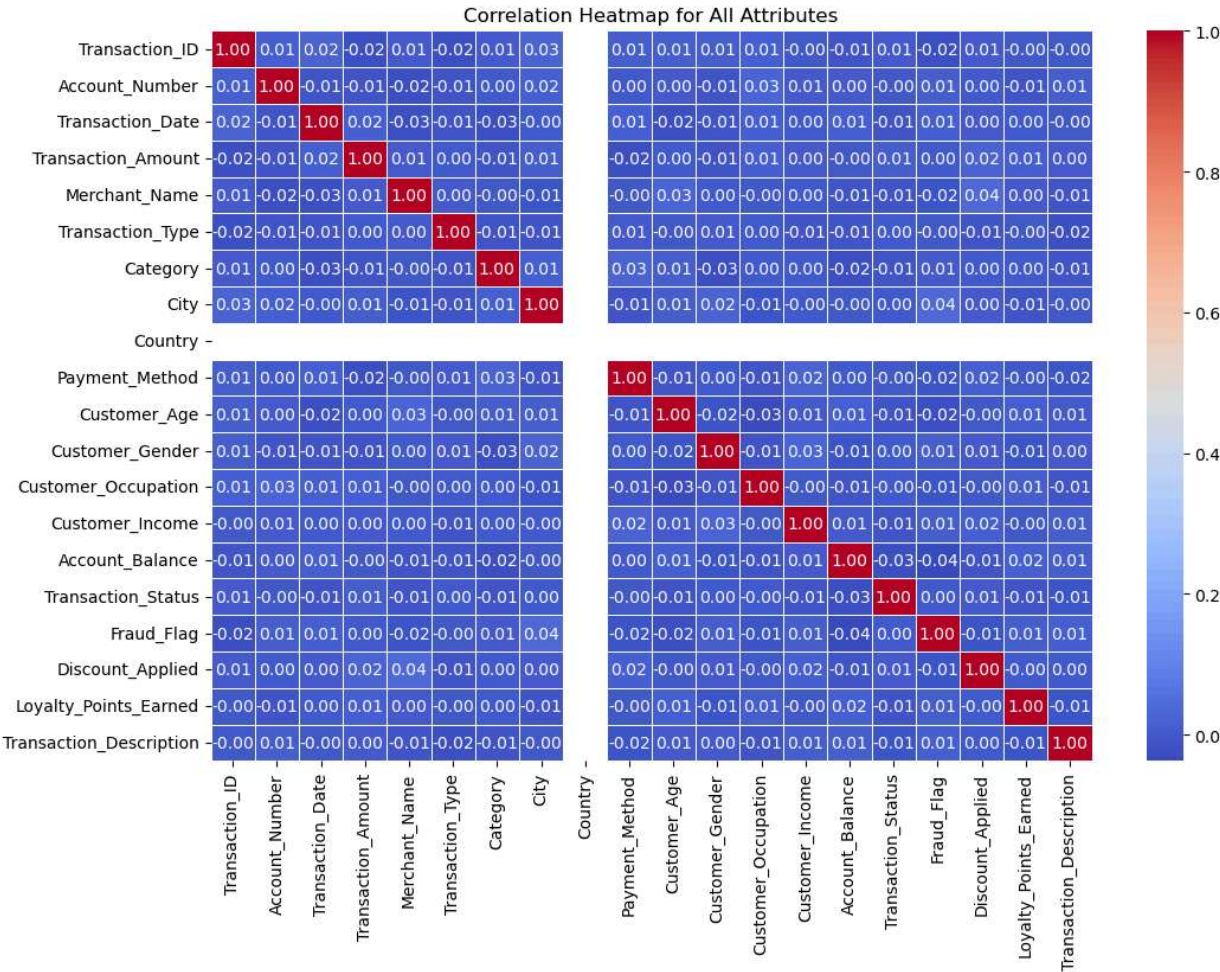
# Step 3: Compute the correlation matrix
correlation_matrix = df.corr()

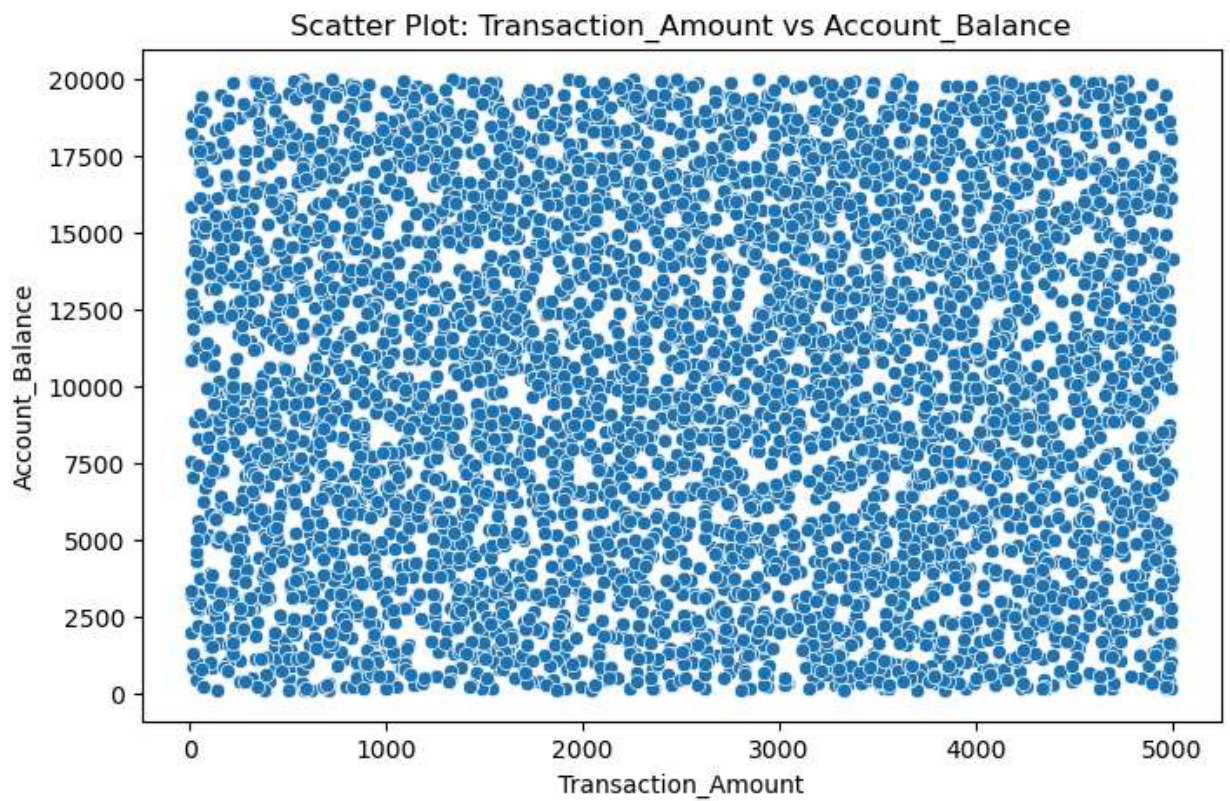
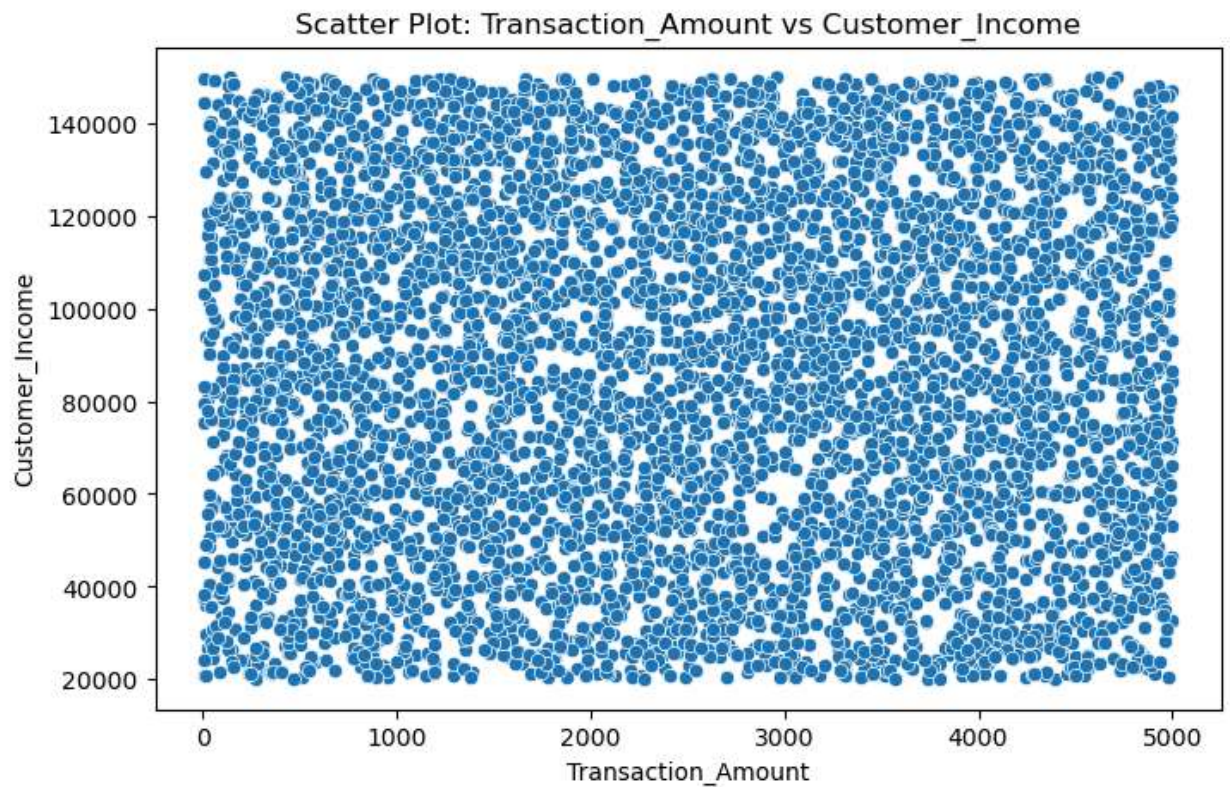
# Step 4: Plot the heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap for All Attributes")
plt.show()

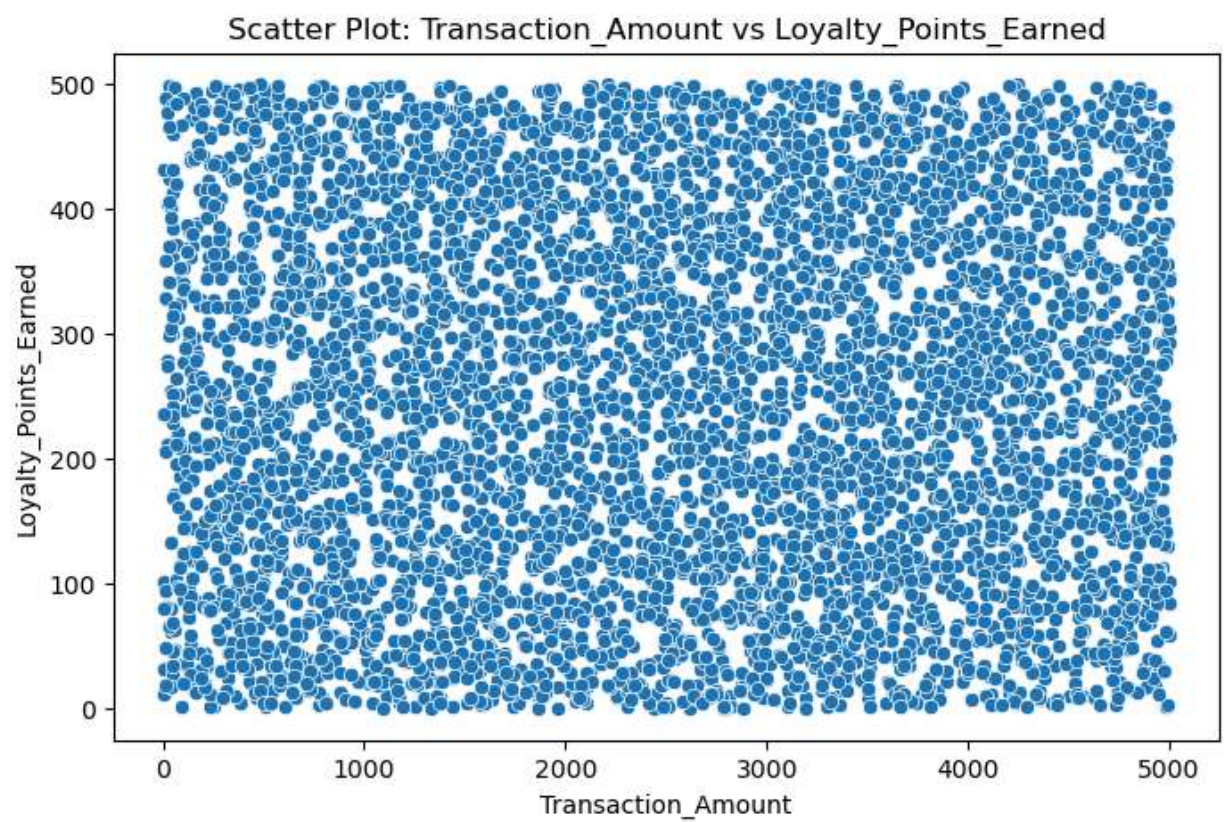
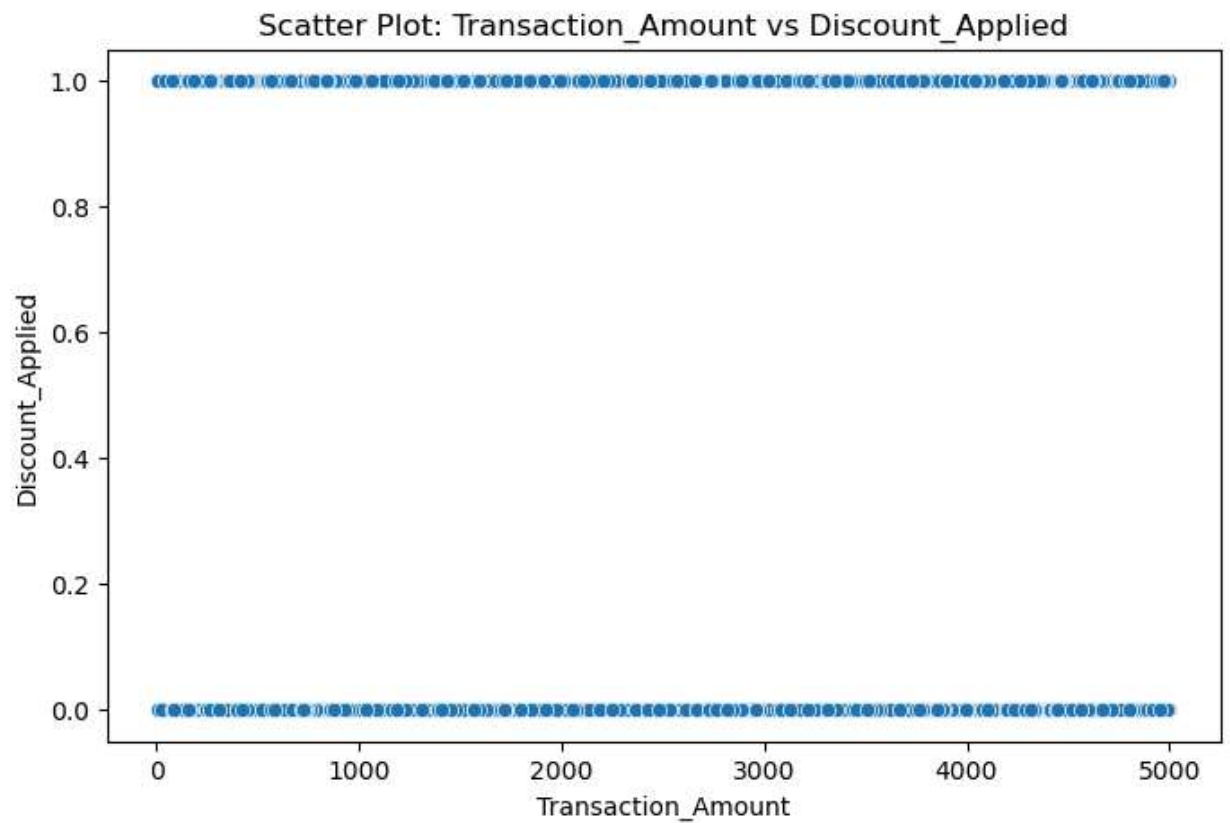
# Step 5: Generate scatter plots for numerical attributes
num_cols = df.select_dtypes(include=['int64', 'float64']).columns
for i in range(len(num_cols)):
    for j in range(i + 1, len(num_cols)):
        plt.figure(figsize=(8, 5))
        sns.scatterplot(x=df[num_cols[i]], y=df[num_cols[j]])
        plt.xlabel(num_cols[i])
        plt.ylabel(num_cols[j])
        plt.title(f"Scatter Plot: {num_cols[i]} vs {num_cols[j]}")
        plt.show()

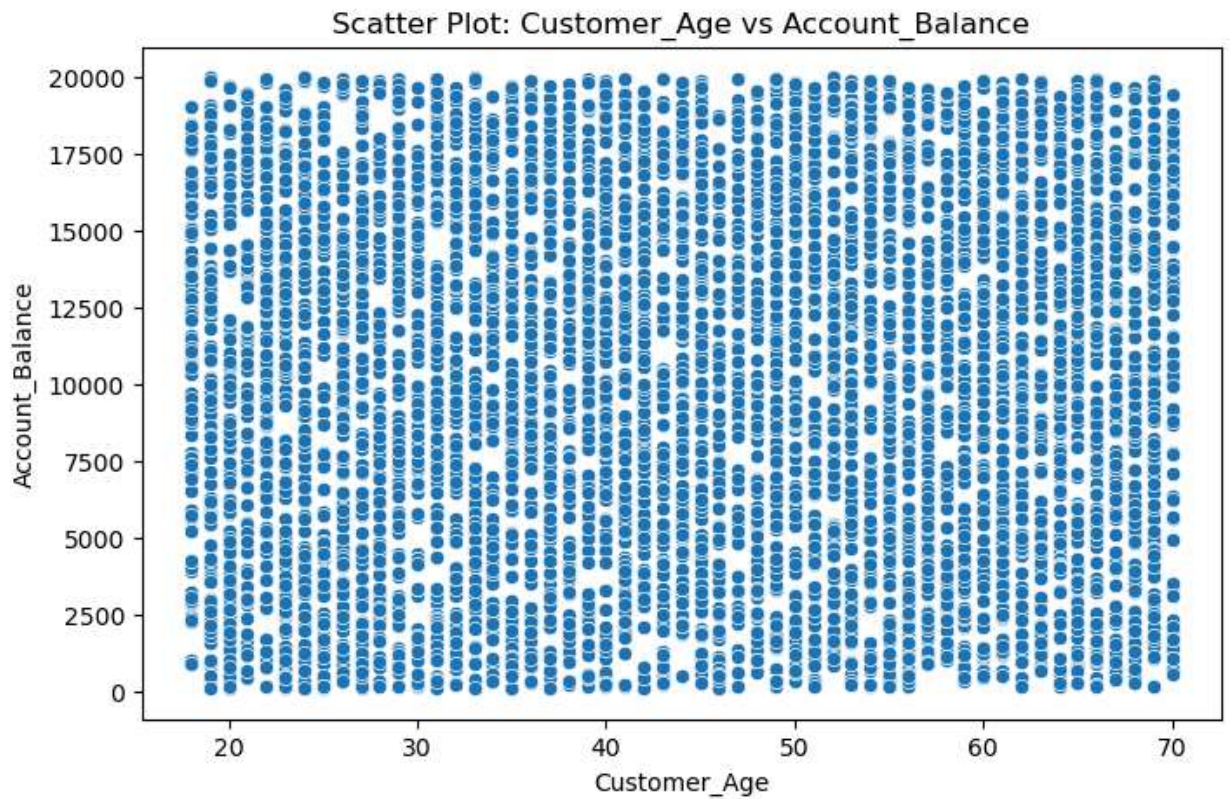
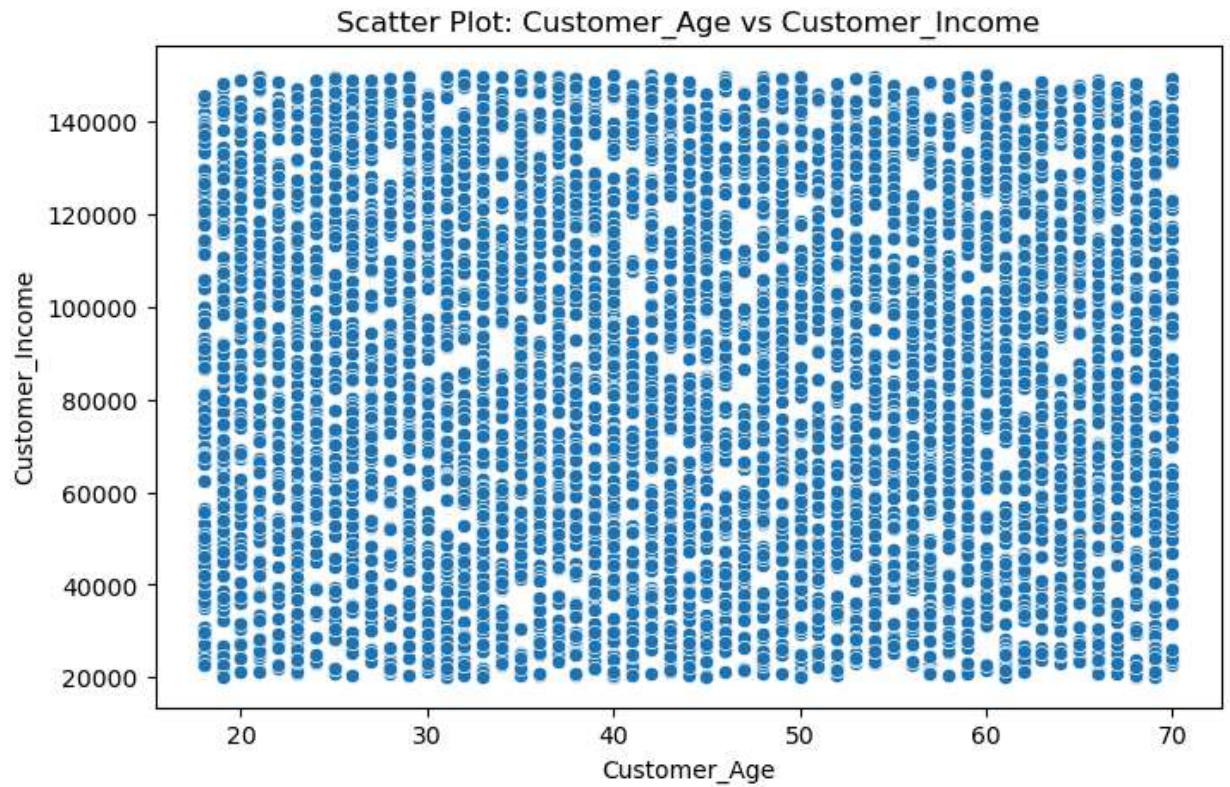
# Step 6: Generate scatter plots for categorical attributes
cat_cols = df.select_dtypes(include=['int64']).columns # Since categorical attributes
for i in range(len(cat_cols)):
    for j in range(i + 1, len(cat_cols)):
        plt.figure(figsize=(8, 5))
        sns.scatterplot(x=df[cat_cols[i]], y=df[cat_cols[j]])
        plt.xlabel(cat_cols[i])
        plt.ylabel(cat_cols[j])
        plt.title(f"Scatter Plot: {cat_cols[i]} vs {cat_cols[j]}")
        plt.show()

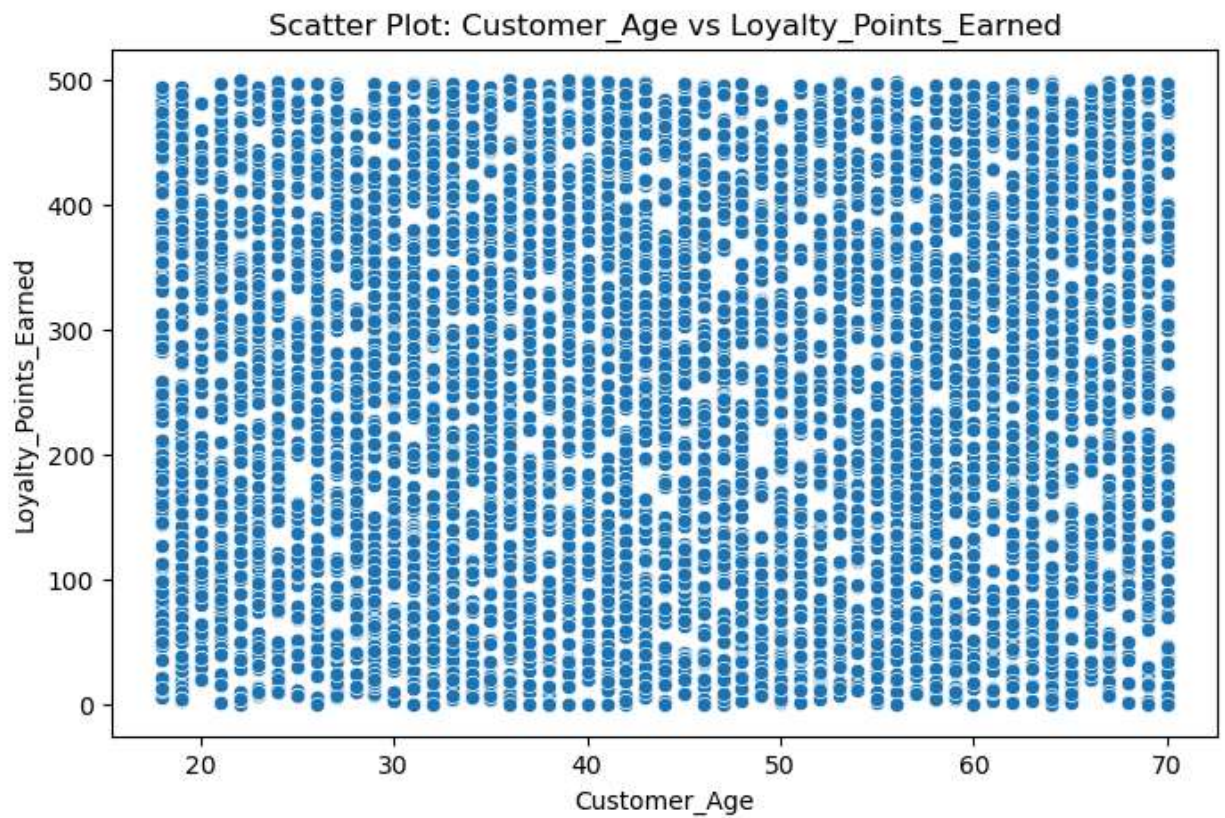
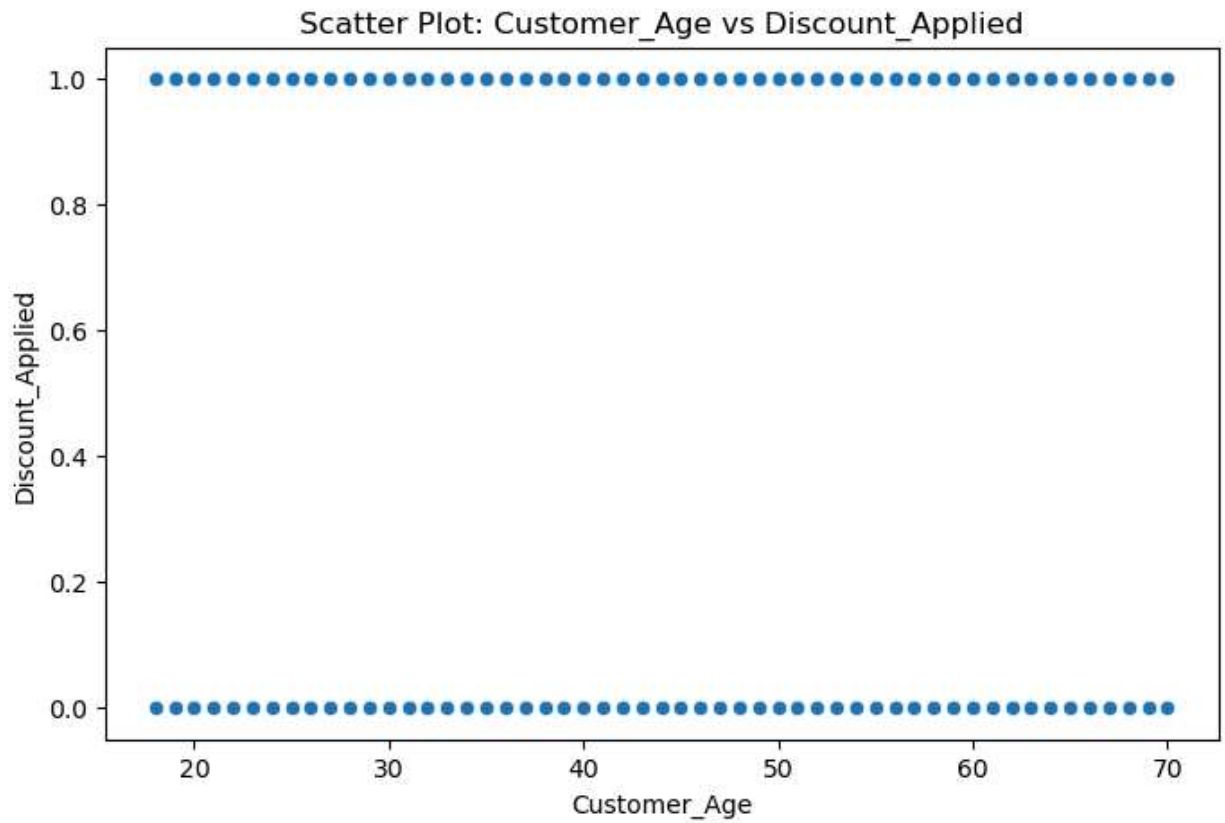
```

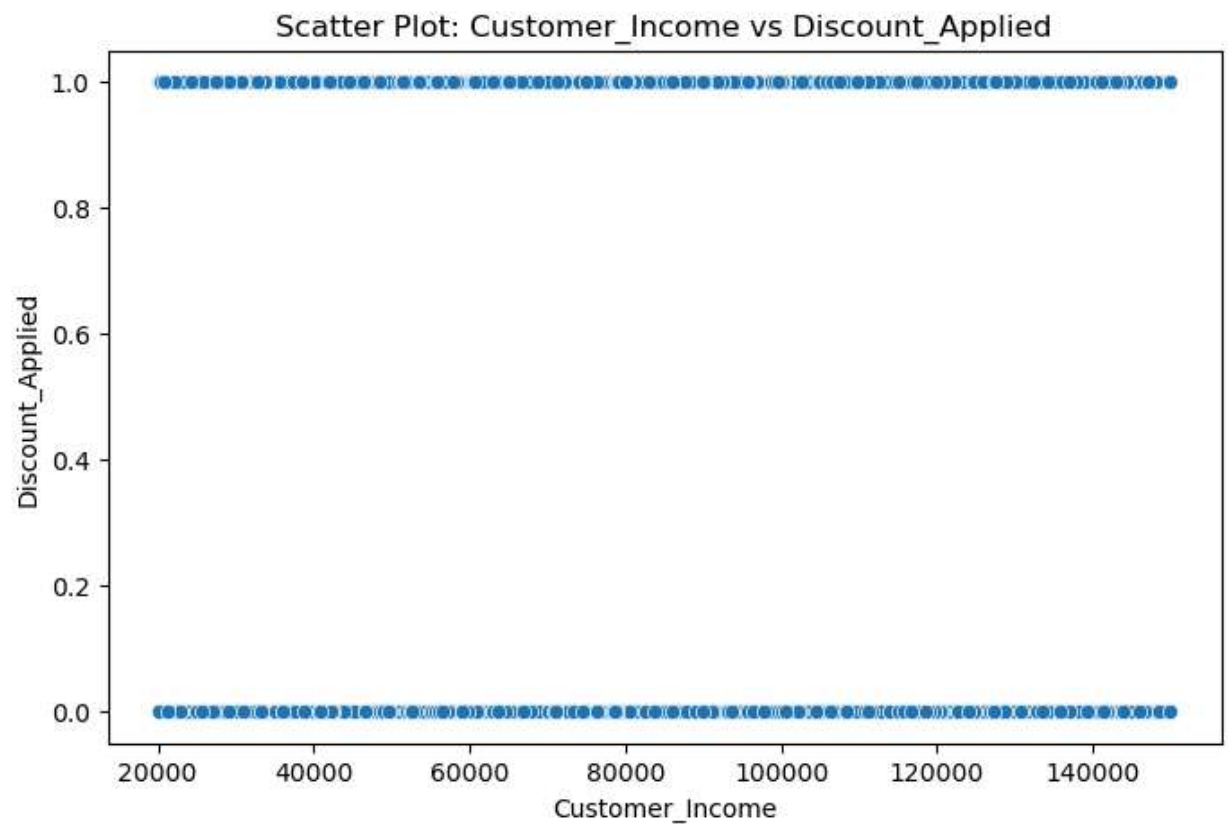
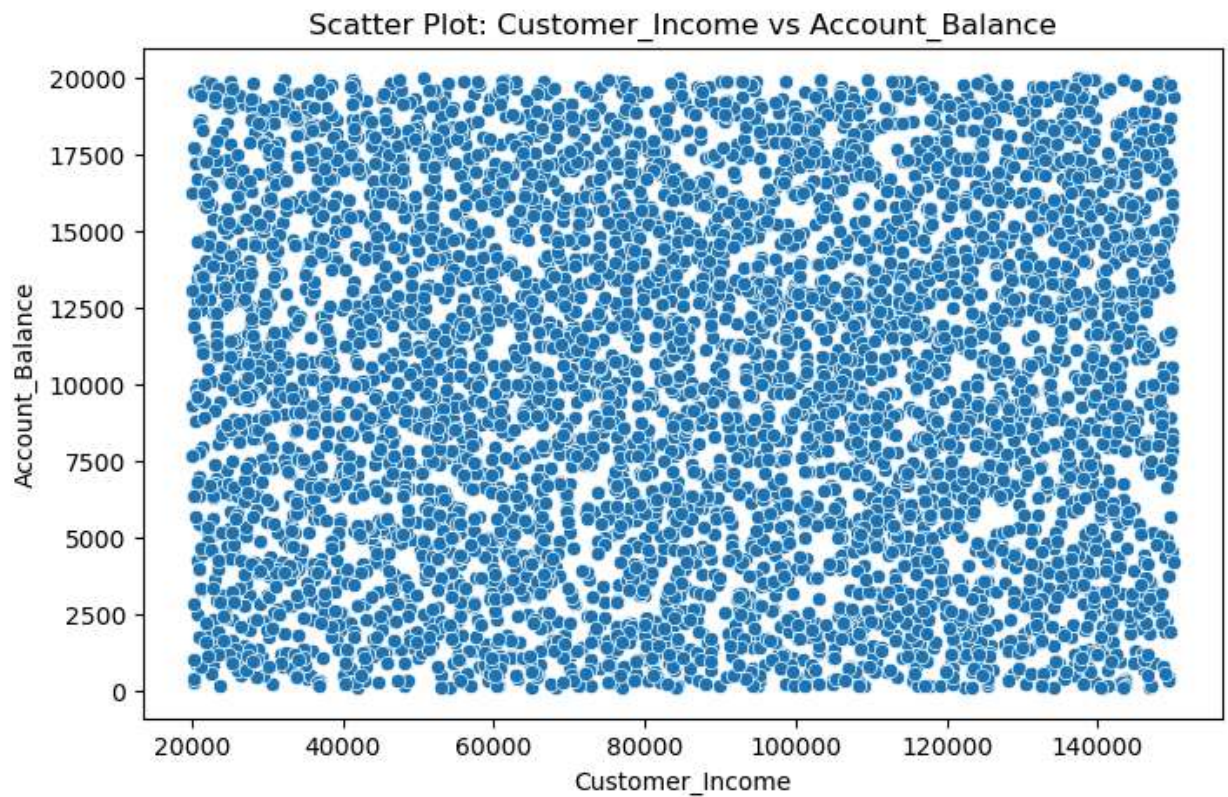



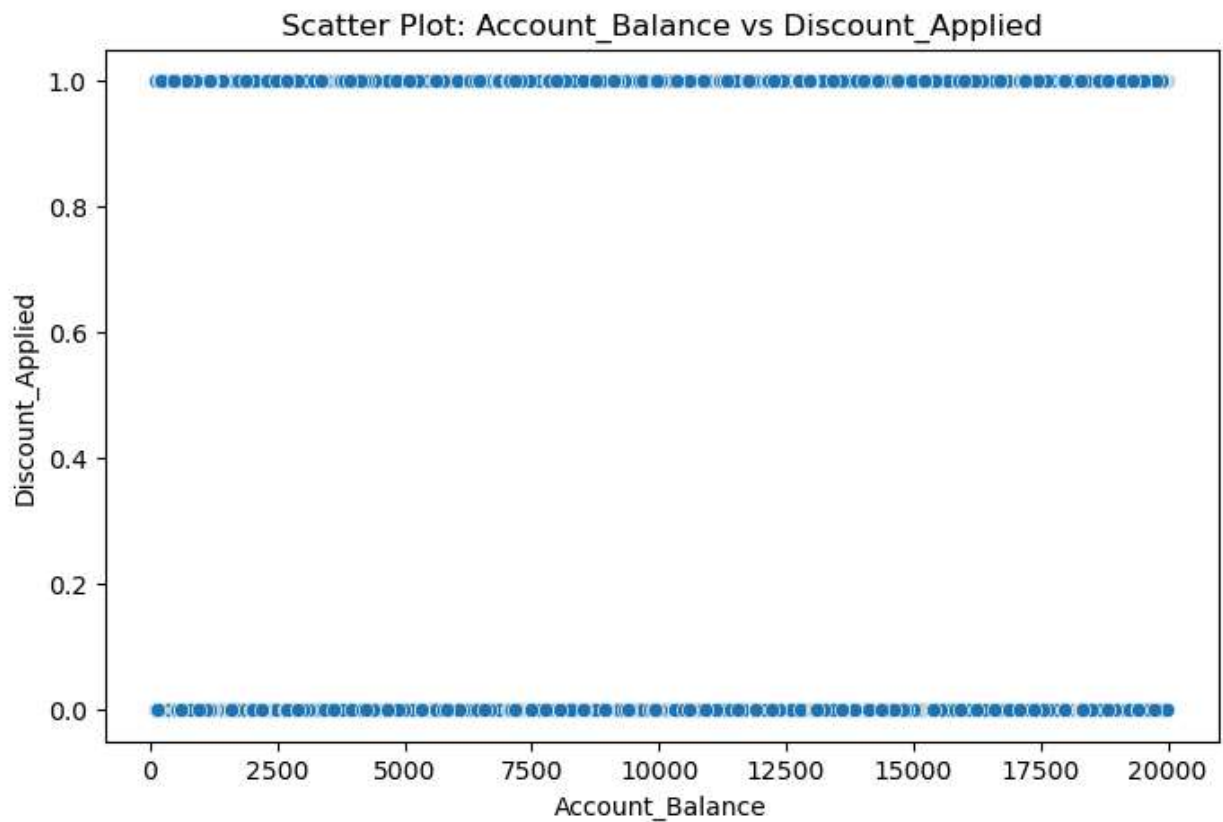
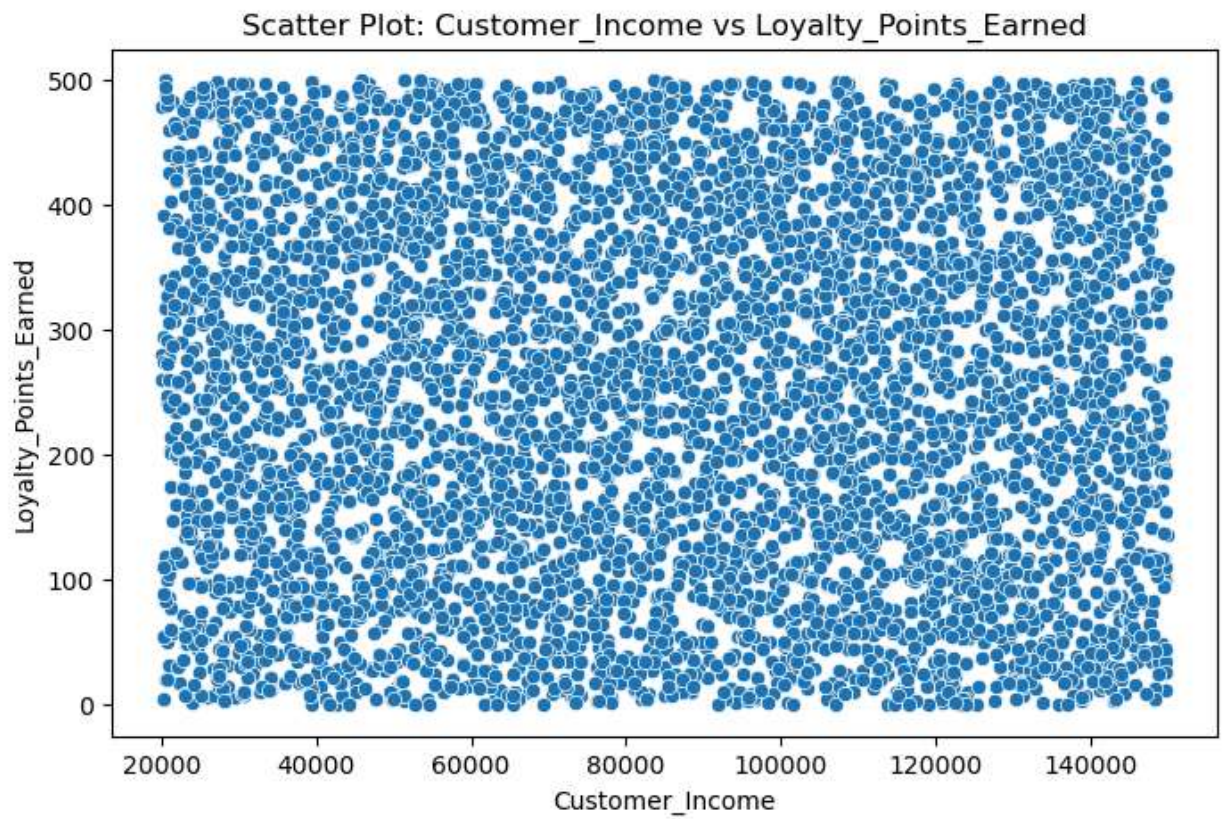


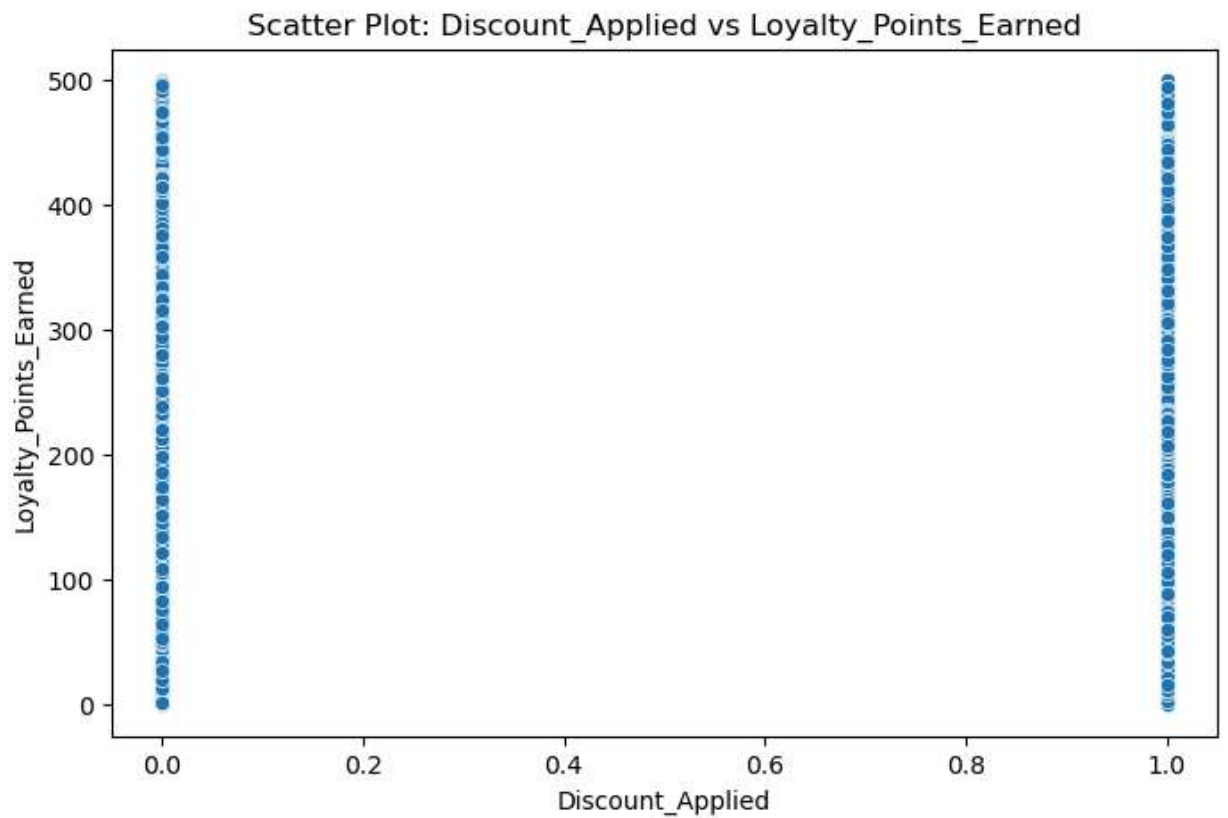
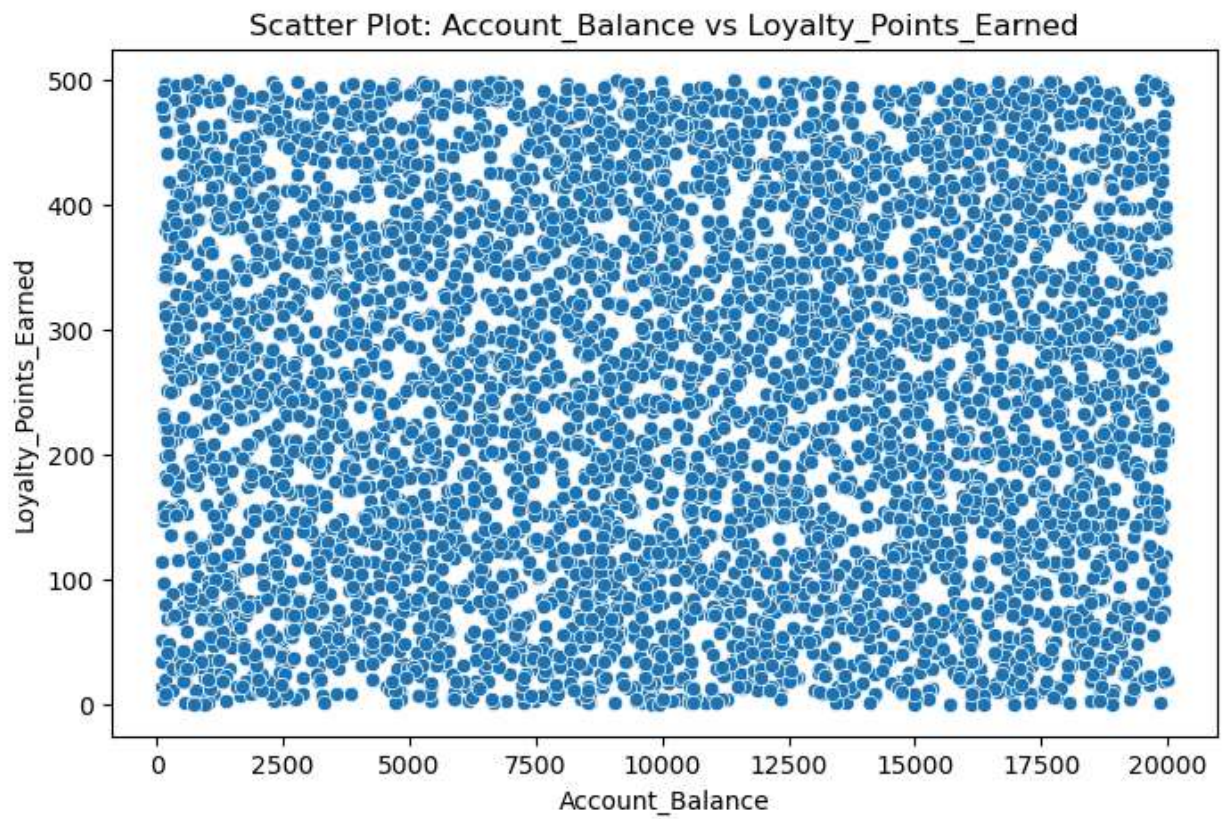


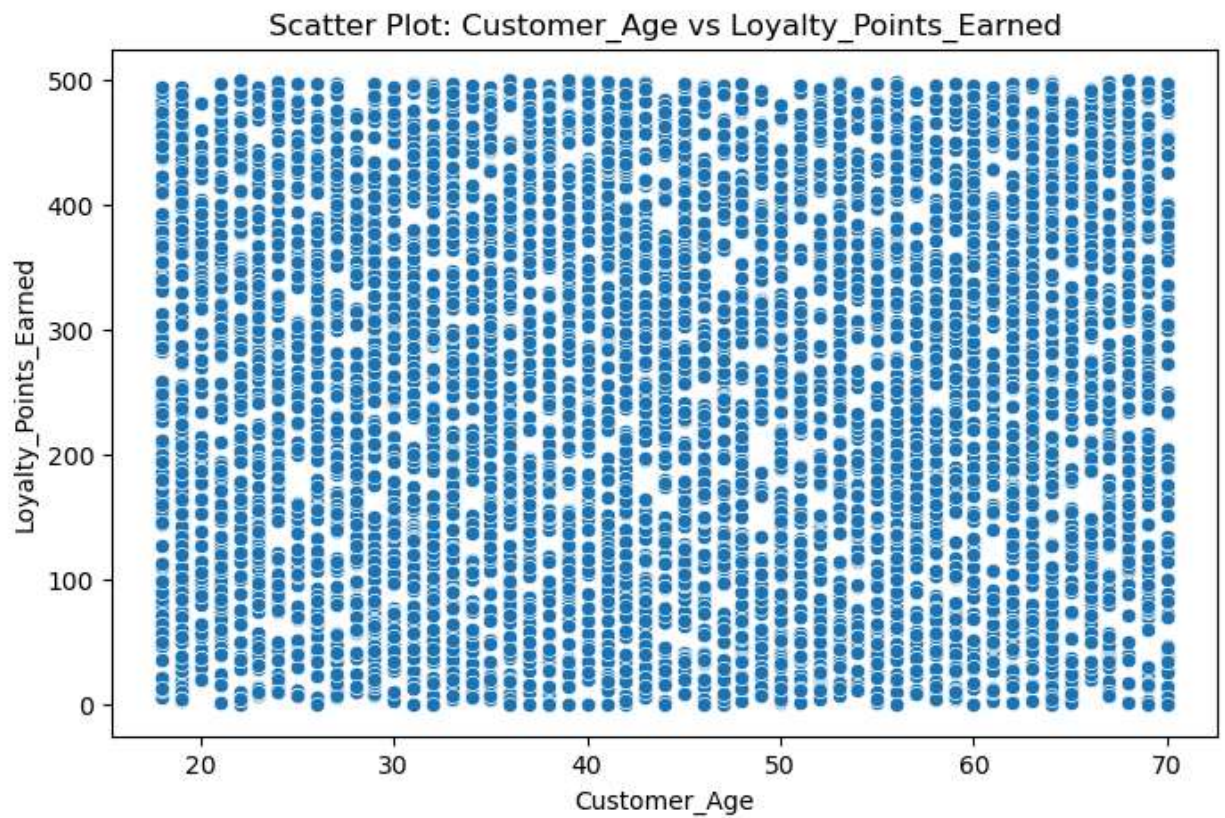
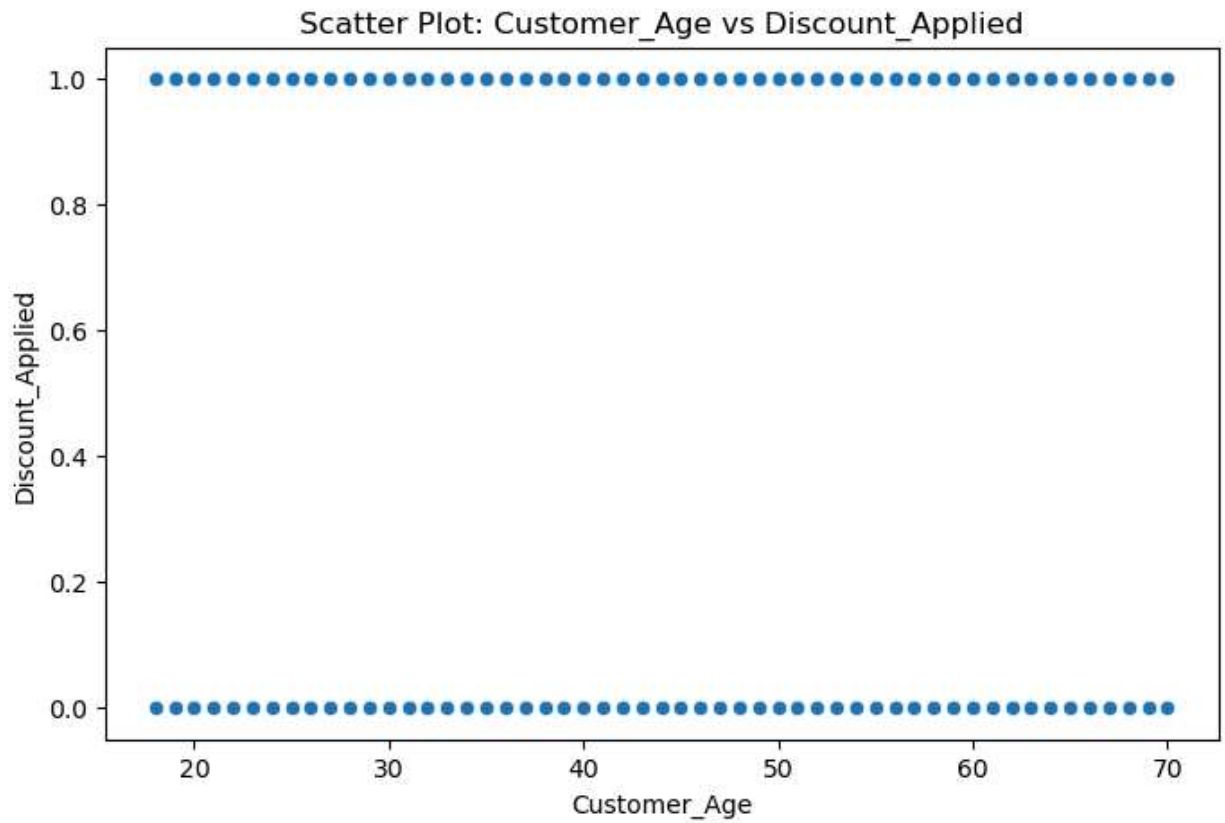


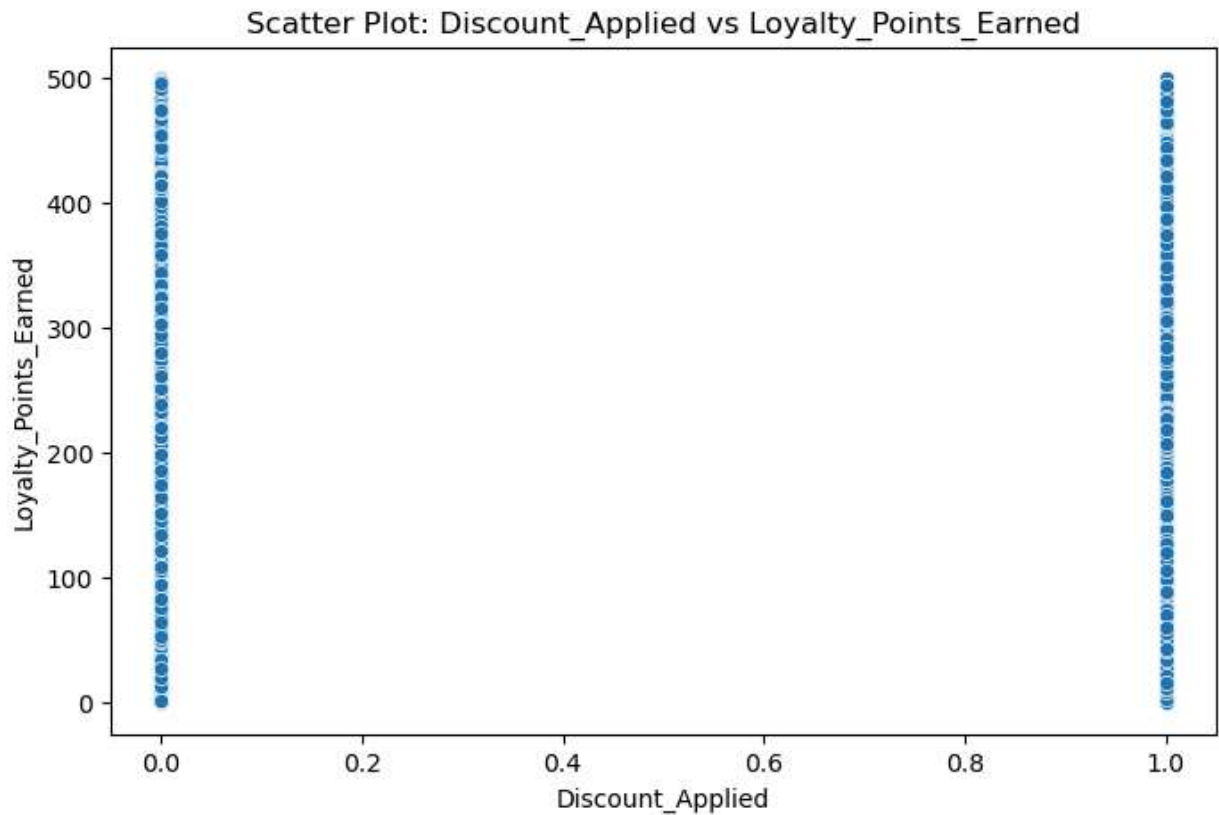












```
In [13]: # Detect and remove outliers using IQR method
Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1

# Filtering out the outliers
df_cleaned = df[~((df < (Q1 - 1.5 * IQR)) | (df > (Q3 + 1.5 * IQR))).any(axis=1)]
```

```
In [14]: print("Original DataFrame shape:", df.shape)
print("Cleaned DataFrame shape:", df_cleaned.shape)
```

```
Original DataFrame shape: (5389, 20)
Cleaned DataFrame shape: (5389, 20)
```

```
In [15]: import matplotlib.pyplot as plt
import seaborn as sns

# Choose a numerical column to visualize
column = 'your_numeric_column_name' # Replace with an actual column name

plt.figure(figsize=(10, 5))

# Boxplot before removing outliers
plt.subplot(1, 2, 1)
sns.boxplot(y=df[column])
plt.title("Before Outlier Removal")

# Boxplot after removing outliers
plt.subplot(1, 2, 2)
sns.boxplot(y=df_cleaned[column])
plt.title("After Outlier Removal")

plt.show()
```



```

-----
KeyError                                Traceback (most recent call last)
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3653, in Index.get_loc(
(self, key)
    3652 try:
-> 3653     return self._engine.get_loc(casted_key)
    3654 except KeyError as err:

File ~\anaconda3\Lib\site-packages\pandas\_libs\index.pyx:147, in pandas._libs.index.
IndexEngine.get_loc()

File ~\anaconda3\Lib\site-packages\pandas\_libs\index.pyx:176, in pandas._libs.index.
IndexEngine.get_loc()

File pandas\_libs\hashtable_class_helper.pxi:7080, in pandas._libs.hashtable.PyObject
HashTable.get_item()

File pandas\_libs\hashtable_class_helper.pxi:7088, in pandas._libs.hashtable.PyObject
HashTable.get_item()

```

KeyError: 'your_numeric_column_name'

The above exception was the direct cause of the following exception:

```

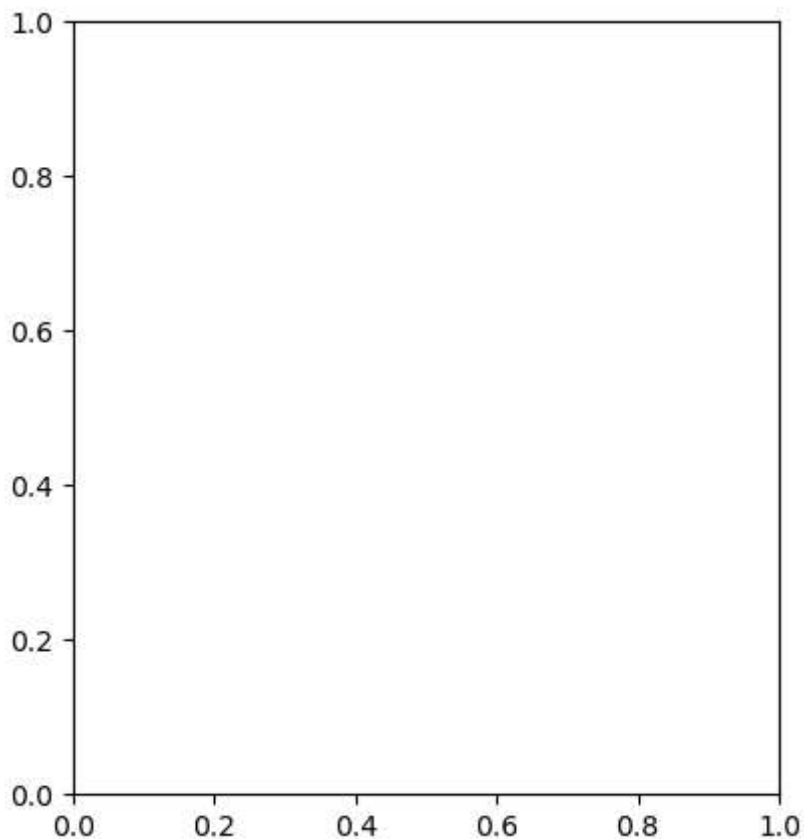
KeyError                                Traceback (most recent call last)
Cell In[15], line 11
     9 # Boxplot before removing outliers
    10 plt.subplot(1, 2, 1)
--> 11 sns.boxplot(y=df[column])
    12 plt.title("Before Outlier Removal")
    14 # Boxplot after removing outliers

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:3761, in DataFrame.__getitem_
__(self, key)
    3759 if self.columns.nlevels > 1:
    3760     return self._getitem_multilevel(key)
-> 3761 indexer = self.columns.get_loc(key)
    3762 if is_integer(indexer):
    3763     indexer = [indexer]

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3655, in Index.get_loc(
(self, key)
    3653     return self._engine.get_loc(casted_key)
    3654 except KeyError as err:
-> 3655     raise KeyError(key) from err
    3656 except TypeError:
    3657     # If we have a listlike key, _check_indexing_error will raise
    3658     # InvalidIndexError. Otherwise we fall through and re-raise
    3659     # the TypeError.
    3660     self._check_indexing_error(key)

```

KeyError: 'your_numeric_column_name'



```
In [16]: print(df.select_dtypes(include=['int64', 'float64']).columns)
Index(['Transaction_Amount', 'Customer_Age', 'Customer_Income',
       'Account_Balance', 'Discount_Applied', 'Loyalty_Points_Earned'],
      dtype='object')
```

```
In [17]: column = "Transaction_Amount" # Replace with your actual column name
```

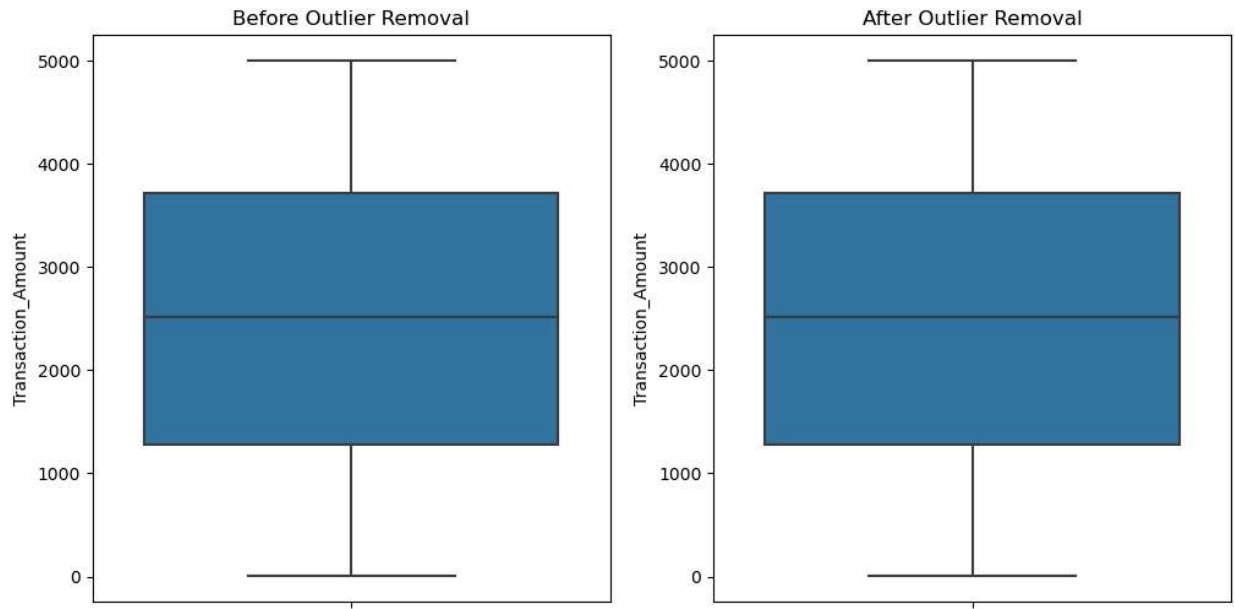
```
In [18]: import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12, 6))

# Boxplot before removing outliers
plt.subplot(1, 2, 1)
sns.boxplot(y=df[column])
plt.title("Before Outlier Removal")

# Boxplot after removing outliers
plt.subplot(1, 2, 2)
sns.boxplot(y=df_cleaned[column])
plt.title("After Outlier Removal")

plt.show()
```

```
In [19]: from scipy import stats
import numpy as np

z_scores = np.abs(stats.zscore(df.select_dtypes(include=['int64', 'float64'])))
df_cleaned = df[(z_scores < 3).all(axis=1)] # Keep rows where all columns have Z-score < 3
print("After Z-score filtering, new shape:", df_cleaned.shape)
```

After Z-score filtering, new shape: (5389, 20)

```
In [20]: # Select only relevant columns (Replace with actual column names)
selected_columns = ["Transaction_Amount", "Transaction_Type", "Customer_Age"]
df_reduced = df[selected_columns]
```

```
In [21]: df_reduced = df.sample(frac=0.3, random_state=42) # Keep 30% of data
```

```
In [22]: correlation_matrix = df.corr().abs()
upper_triangle = correlation_matrix.where(
    np.triu(np.ones(correlation_matrix.shape), k=1).astype(bool)
)

# Drop columns with high correlation
to_drop = [column for column in upper_triangle.columns if any(upper_triangle[column] > 0.9)]
df_reduced = df.drop(columns=to_drop)
```

```
In [23]: df['Transaction_Date'] = pd.to_datetime(df['Transaction_Date'])
df_reduced = df[df['Transaction_Date'] > '2023-01-01'] # Keep transactions after Jan 1, 2023
```

```
In [24]: print("Original Dataset Shape:", df.shape)
print("Reduced Dataset Shape:", df_reduced.shape)
```

Original Dataset Shape: (5389, 20)
Reduced Dataset Shape: (0, 20)

```
In [25]: df_reduced = df.sample(frac=0.3, random_state=42) # Keep 30% of data
```

```
In [26]: df['Transaction_Date'] = pd.to_datetime(df['Transaction_Date'], errors='coerce')
df_reduced = df[df['Transaction_Date'] > '2023-01-01']
```

```
In [27]: if df_reduced.shape[1] == 0: # If no columns left
         df_reduced = df.copy() # Restore original data
```

```
In [28]: print(df_reduced.head()) # Show first 5 rows
         print(df_reduced.shape) # Show dataset size
```

Empty DataFrame

Columns: [Transaction_ID, Account_Number, Transaction_Date, Transaction_Amount, Merchant_Name, Transaction_Type, Category, City, Country, Payment_Method, Customer_Age, Customer_Gender, Customer_Occupation, Customer_Income, Account_Balance, Transaction_Status, Fraud_Flag, Discount_Applied, Loyalty_Points_Earned, Transaction_Description]
Index: []
(0, 20)

```
In [29]: print("Original Dataset Shape:", df.shape)
         print(df.head()) # See the first 5 rows
```

Original Dataset Shape: (5389, 20)

	Transaction_ID	Account_Number	Transaction_Date	\
0	4047	1777	1970-01-01 00:00:00.000000912	
1	771	266	1970-01-01 00:00:00.000003600	
2	4039	4268	1970-01-01 00:00:00.000002967	
3	3218	4831	1970-01-01 00:00:00.000004704	
4	490	2124	1970-01-01 00:00:00.000004491	

	Transaction_Amount	Merchant_Name	Transaction_Type	Category	City	\
0	3198.94	1994	1	11	6	
1	129.93	113	0	6	5	
2	1378.77	2135	0	10	4	
3	1119.94	3106	0	7	1	
4	3683.67	607	0	3	9	

	Country	Payment_Method	Customer_Age	Customer_Gender	\
0	0	4	55	2	
1	0	2	26	2	
2	0	2	29	2	
3	0	4	60	1	
4	0	3	29	2	

	Customer_Occupation	Customer_Income	Account_Balance	Transaction_Status	\
0	490	80466.03	350.28	0	
1	105	145574.25	9797.81	1	
2	439	33447.18	12399.85	0	
3	498	108801.45	16057.64	0	
4	171	100985.12	14940.54	0	

	Fraud_Flag	Discount_Applied	Loyalty_Points_Earned	\
0	0	1	304	
1	1	0	383	
2	1	0	497	
3	1	1	495	
4	1	1	292	

	Transaction_Description
0	3810
1	276
2	1776
3	3201
4	1545


```
In [30]: print(df['Transaction_Date'].head()) # See the first few dates
print(df['Transaction_Date'].dtype) # Check the data type
```

```
0    1970-01-01 00:00:00.000000912
1    1970-01-01 00:00:00.000003600
2    1970-01-01 00:00:00.000002967
3    1970-01-01 00:00:00.000004704
4    1970-01-01 00:00:00.000004491
Name: Transaction_Date, dtype: datetime64[ns]
datetime64[ns]
```

```
In [31]: df_reduced = df[df['Transaction_Date'] > '2023-01-01']
print(df_reduced.shape) # Check if it's still empty
```

```
(0, 20)
```

```
In [32]: df_reduced = df.sample(frac=0.3, random_state=42)
print(df_reduced.shape) # Should not be (0, 20)
```

```
(1617, 20)
```

```
In [33]: print("Columns before reduction:", df.shape[1])
df_reduced = df.drop(columns=to_drop, errors='ignore')
print("Columns after reduction:", df_reduced.shape[1])
```

```
Columns before reduction: 20
Columns after reduction: 20
```

```
In [34]: df_reduced = df.copy()
print(df_reduced.shape) # Should match the original dataset
```

```
(5389, 20)
```

```
In [1]: categorical_columns = df.select_dtypes(include=['object', 'category', 'bool']).columns

for col in categorical_columns:
    print(f"Column: {col} - Unique Classes: {df[col].unique()}")
    print(df[col].unique()) # Display the actual classes
    print("-" * 40)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[1], line 1
----> 1 categorical_columns = df.select_dtypes(include=['object', 'category', 'boo
1']).columns
      3 for col in categorical_columns:
      4     print(f"Column: {col} - Unique Classes: {df[col].unique()}")

NameError: name 'df' is not defined
```

```
In [3]: import pandas as pd

# Step 1: Load the dataset
file_path = r"C:\Users\hethv\OneDrive\Desktop\Banking_Transactions_USA_2023_2024.csv"
df = pd.read_csv(file_path)

# Step 2: Identify Categorical Columns
categorical_columns = df.select_dtypes(include=['object', 'category', 'bool']).columns

# Step 3: Display Unique Classes for Each Categorical Column
for col in categorical_columns:
```

```
print(f"Column: {col} - Unique Classes: {df[col].nunique()}")  
print(df[col].unique()) # Display the actual class values  
print("-" * 40)
```


Column: Transaction_ID - Unique Classes: 5389

['bdd640fb-0667-4ad1-9c80-317fa3b1799d'
'23b8c1e9-3924-46de-beb1-3b9046685257'
'bd9c66b3-ad3c-4d6d-9a3d-1fa7bc8960a9' ...
'ea725584-5e6e-487a-a939-f79018177e4a'
'03539b9a-eb9b-4d98-80b7-10a640f66fde'
'a5b01639-0895-4db1-85e5-26e19b559178']

Column: Account_Number - Unique Classes: 5389

['IUPM04409079772781' 'BLAT22216107051843' 'UTXA55295806601382' ...
'DOBF62458522431672' 'BRHX88846108622659' 'CTJA66942505562975']

Column: Transaction_Date - Unique Classes: 5376

['05-11-2023 15:54' '21-04-2024 22:21' '17-07-2023 13:25' ...
'30-11-2024 00:58' '09-08-2023 00:31' '16-03-2024 09:44']

Column: Merchant_Name - Unique Classes: 4880

['Houston Group' 'Anderson-Phillips' 'Jensen Group' ... 'Banks-Reed'
'Townsend-Bartlett' 'Molina, Smith and King']

Column: Transaction_Type - Unique Classes: 2

['Debit' 'Credit']

Column: Category - Unique Classes: 14

['Transport' 'Grocery' 'Shopping' 'Healthcare' 'Entertainment' 'Savings'
'Clothing' 'Housing' 'Electronics' 'Education' 'Utilities' 'Fitness'
'Travel' 'Food']

Column: City - Unique Classes: 10

['Phoenix' 'Philadelphia' 'New York' 'Dallas' 'San Jose' 'Los Angeles'
'San Diego' 'San Antonio' 'Chicago' 'Houston']

Column: Country - Unique Classes: 1

['USA']

Column: Payment_Method - Unique Classes: 5

['Online Transfer' 'Debit Card' 'E-Wallet' 'Credit Card' 'Cash']

Column: Customer_Gender - Unique Classes: 3

['Others' 'Male' 'Female']

Column: Customer_Occupation - Unique Classes: 639

['Quality manager' 'Civil Service fast streamer' 'Pilot, airline'
'Radiographer, therapeutic' 'Diplomatic Services operational officer'
'Osteopath' 'Event organiser' 'Financial controller'
'Copywriter, advertising' 'Hydrologist' 'Special effects artist'
'Designer, television/film set' 'Radio producer' 'Conservator, furniture'
'Communications engineer' 'Animal nutritionist'
'Lecturer, further education' 'Scientist, research (medical)'
'Nurse, mental health' 'Sound technician, broadcasting/film/video'
'Engineer, biomedical' 'Artist' 'Restaurant manager, fast food'
'Conservation officer, historic buildings' 'Nutritional therapist'
'Animal technologist' 'Child psychotherapist' 'Local government officer'
'Maintenance engineer' 'Media planner' 'Analytical chemist'
'Chartered management accountant' 'Medical laboratory scientific officer'
'IT trainer' 'Engineer, mining' 'Sales professional, IT'
'Intelligence analyst' 'Engineer, maintenance (IT)'
'Clinical psychologist' 'Warden/ranger' 'Engineer, agricultural'
'Immunologist' 'Multimedia programmer' 'Information systems manager'
'Regulatory affairs officer' 'Multimedia specialist']

'Waste management officer' 'Optician, dispensing' 'Financial planner'
 'Haematologist' 'Community education officer' 'Probation officer'
 'Electrical engineer' 'Clinical biochemist' 'Town planner'
 'Buyer, retail' 'Lecturer, higher education'
 'Social research officer, government' 'Private music teacher'
 'Training and development officer' 'Education officer, community'
 'Accountant, chartered public finance' 'Administrator, education'
 'Surveyor, building' 'Politician's assistant' 'Retail banker'
 'Production assistant, television' 'Development worker, community'
 'Food technologist' 'Community pharmacist'
 'Tourist information centre manager' 'Gaffer' 'Community arts worker'
 'Technical sales engineer' 'Psychiatrist' 'Engineer, structural'
 'Press photographer' 'Ecologist' 'Production manager' 'Charity officer'
 'Dentist' 'Building surveyor' 'Field seismologist' 'Public librarian'
 'Civil engineer, consulting' 'Hydrogeologist' 'Midwife' 'Tour manager'
 'Video editor' 'Engineer, control and instrumentation' 'Curator'
 'Proofreader' 'TEFL teacher' 'Retail buyer'
 'Interior and spatial designer' 'Textile designer'
 'Higher education careers adviser' 'Operational researcher'
 'Biomedical scientist' 'Doctor, general practice' 'Broadcast journalist'
 'Learning mentor' 'Paediatric nurse' 'Theatre manager'
 'Clinical cytogeneticist'
 'Armed forces logistics/support/administrative officer'
 'Surveyor, hydrographic' 'Chiropractor' 'Pharmacologist' 'Health visitor'
 'Veterinary surgeon' 'Operational investment banker' 'Charity fundraiser'
 'Teacher, special educational needs' 'Pensions consultant' 'Cabin crew'
 'Hospital pharmacist' 'Arboriculturist' 'Industrial buyer'
 'IT consultant' 'Museum/gallery curator' 'Risk manager'
 'Chemical engineer' 'Legal secretary'
 'Programme researcher, broadcasting/film/video'
 'Clinical research associate' 'Learning disability nurse'
 'Control and instrumentation engineer' 'Scientist, forensic'
 'Engineer, land' 'Database administrator' 'Physiological scientist'
 'Herpetologist' 'Designer, graphic' 'Podiatrist'
 'Designer, ceramics/pottery' 'Education officer, museum'
 'Museum/gallery exhibitions officer' 'Accountant, chartered management'
 'Amenity horticulturist' 'Therapist, nutritional'
 'Building control surveyor' 'Games developer'
 'Civil engineer, contracting' 'Economist' 'Magazine features editor'
 'Police officer' 'Engineer, automotive' 'Designer, furniture'
 'Accountant, chartered certified' 'Chemist, analytical'
 'Adult guidance worker' 'Armed forces technical officer'
 'Therapist, music' 'Health and safety inspector'
 'Land/geomatics surveyor' 'Museum education officer'
 'Recruitment consultant' 'Network engineer' 'Medical illustrator'
 'Television production assistant' 'Visual merchandiser'
 'Occupational psychologist' 'Ophthalmologist'
 'Research scientist (medical)' 'Geophysicist/field seismologist'
 'Nurse, children's' 'Energy engineer' 'Trading standards officer'
 'Prison officer' 'Presenter, broadcasting'
 'Scientific laboratory technician' 'Tax inspector'
 'Consulting civil engineer' 'Loss adjuster, chartered'
 'Psychiatric nurse' 'English as a foreign language teacher'
 'Acupuncturist' 'Radiation protection practitioner' 'Catering manager'
 'Research officer, political party' 'Surveyor, minerals'
 'Materials engineer' 'Equality and diversity officer' 'Make'
 'Nurse, adult' 'Financial risk analyst' 'Engineer, energy'
 'Administrator, arts' 'Toxicologist' 'Risk analyst' 'Archaeologist'
 'Electronics engineer' 'Musician' 'Chief Financial Officer'
 'Sports development officer' 'Applications developer' 'Water engineer'

'Systems developer' 'Banker' 'Jewellery designer' 'Interpreter'
 'Surveyor, land/geomatics' 'Tourism officer' 'Youth worker'
 'Manufacturing engineer' 'Emergency planning/management officer'
 'Secondary school teacher' 'Education administrator' 'Chief of Staff'
 'Airline pilot' 'Runner, broadcasting/film/video' 'Contractor'
 'Scientist, clinical (histocompatibility and immunogenetics)'
 'Engineer, site' 'Administrator, charities/voluntary organisations'
 'Advice worker' 'Outdoor activities/education manager'
 'Forensic scientist' 'Psychologist, counselling'
 'Geophysical data processor' 'Plant breeder/geneticist'
 'Rural practice surveyor' 'Camera operator' 'Corporate investment banker'
 'Dancer' 'Records manager' 'Theme park manager' 'Psychologist, forensic'
 'Colour technologist' 'Architectural technologist' 'Barista'
 'Therapist, speech and language' 'Facilities manager'
 'Geographical information systems officer' 'Public affairs consultant'
 'Investment banker, corporate' 'Dispensing optician' 'Financial manager'
 'Designer, interior/spatial' 'Solicitor' 'Editor, film/video'
 'Equities trader' 'Engineer, civil (consulting)' 'Music therapist'
 'Merchant navy officer' 'Race relations officer' 'Careers adviser'
 'Graphic designer' 'Lobbyist' 'Solicitor, Scotland' 'Purchasing manager'
 'Civil Service administrator' 'Community development worker'
 'Environmental consultant' 'Teacher, music' 'Copy' 'Illustrator'
 'Scientist, audiological' 'Photographer'
 'Special educational needs teacher' 'Product designer'
 'Glass blower/designer' 'Furniture conservator/restorer'
 'Journalist, magazine' 'Best boy' 'International aid/development worker'
 'Occupational hygienist' 'Horticulturist, commercial'
 'Automotive engineer' 'Print production planner' 'Brewing technologist'
 'Environmental health practitioner' 'Public relations officer'
 'Scientist, biomedical' 'Energy manager' 'Editor, commissioning'
 'Landscape architect' 'Licensed conveyancer' 'Ergonomist'
 'Accommodation manager' 'Museum/gallery conservator'
 'Radiographer, diagnostic' 'Administrator, Civil Service'
 'Librarian, public' 'Professor Emeritus'
 'Chartered legal executive (England and Wales)'
 'Surveyor, commercial/residential' 'Pharmacist, hospital'
 'Scientist, marine' 'Therapist, sports'
 'Sales promotion account executive' 'Recycling officer'
 'Diagnostic radiographer' 'Contracting civil engineer'
 'Educational psychologist' 'Phytotherapist' 'Therapeutic radiographer'
 'Chiropodist' 'Cytogeneticist' 'Retail merchandiser'
 'Senior tax professional/tax inspector' 'Software engineer'
 'Drilling engineer' 'Futures trader' 'Art therapist'
 'Engineer, manufacturing systems' 'Film/video editor' 'Neurosurgeon'
 'Designer, jewellery' 'Dealer' 'Exhibitions officer, museum/gallery'
 'Environmental manager' 'Trade mark attorney'
 'Conservation officer, nature' 'Scientist, research (physical sciences)'
 'Engineer, electronics' 'Holiday representative'
 'Teacher, English as a foreign language' 'IT sales professional'
 'Engineer, aeronautical' 'Sports administrator' 'Operations geologist'
 'Counselling psychologist' 'Writer' 'Television floor manager'
 'Therapist, occupational' 'Data processing manager'
 'Health promotion specialist' 'Designer, blown glass/stained glass'
 'Web designer' 'Investment banker, operational'
 'Television/film/video producer' 'Teacher, secondary school'
 'Surveyor, rural practice' 'Investment analyst' 'Metallurgist'
 'Claims inspector/assessor' 'Data scientist' 'Estate agent'
 'Newspaper journalist' 'Research officer, government'
 'Occupational therapist' 'Engineer, petroleum'
 'Fast food restaurant manager' 'Doctor, hospital'

'Estate manager/land agent' 'Fine artist'
 'Chartered certified accountant' 'Psychologist, clinical' 'Ship broker'
 'Herbalist' 'Advertising account executive' 'Teacher, adult education'
 'Environmental education officer' 'Fish farm manager'
 'Speech and language therapist' 'Logistics and distribution manager'
 'Theatre stage manager' 'Passenger transport manager' 'Producer, radio'
 'Architect' 'Financial trader' 'Patent examiner'
 'Historic buildings inspector/conservation officer' 'Office manager'
 'Animator' 'Sports coach' 'Psychologist, educational'
 'Publishing rights manager' 'Marketing executive' 'Optometrist'
 'Further education lecturer' 'Geneticist, molecular'
 'Surveyor, building control' 'Production assistant, radio'
 'Sport and exercise psychologist' 'Telecommunications researcher'
 'Travel agency manager' 'Scientist, research (life sciences)'
 'Pension scheme manager' 'Farm manager' 'Teacher, primary school'
 'Armed forces training and education officer' 'Administrator'
 'Technical author' 'Forest/woodland manager' 'Seismic interpreter'
 'Forensic psychologist' 'Minerals surveyor' 'Barrister'
 'Restaurant manager' 'Engineer, broadcasting (operations)'
 'Marine scientist' 'Programmer, multimedia' 'Sub' 'Mining engineer'
 'General practice doctor' 'Agricultural consultant'
 'Research scientist (maths)' 'Broadcast presenter'
 'Lighting technician, broadcasting/film/video' 'Insurance broker'
 'Building services engineer' 'Commercial art gallery manager'
 'Health and safety adviser' 'Public house manager' 'Press sub'
 'Pathologist' 'Corporate treasurer' 'Health physicist'
 'Volunteer coordinator' 'Publishing copy' 'Adult nurse'
 'Insurance claims handler' 'Horticultural consultant'
 'Production designer, theatre/television/film' 'Medical secretary'
 'Surveyor, mining' 'Journalist, newspaper' 'Surveyor, quantity'
 'Aeronautical engineer' 'Systems analyst'
 'Education officer, environmental' 'Mudlogger' 'Air broker'
 'Insurance underwriter' 'Arts administrator' 'Lawyer' 'Lexicographer'
 'Teacher, early years/pre' 'Legal executive' 'Site engineer'
 'Naval architect' 'Geochemist' 'Engineer, drilling' 'Translator'
 'Conservator, museum/gallery' 'Engineering geologist'
 'Psychotherapist, child' 'Statistician' 'Medical technical officer'
 'Clinical scientist, histocompatibility and immunogenetics' 'Archivist'
 'Magazine journalist' 'Company secretary' 'Chief Marketing Officer'
 'Chief Operating Officer' 'Engineer, materials' 'Set designer'
 'Medical physicist' 'Engineer, civil (contracting)'
 'Housing manager/officer' 'Merchandiser, retail'
 'Garment/textile technologist' 'Manufacturing systems engineer'
 'Dance movement psychotherapist' 'Customer service manager'
 'Dramatherapist' 'Biomedical engineer'
 'Research scientist (life sciences)' 'Buyer, industrial'
 'Wellsite geologist' 'Product/process development scientist'
 'Firefighter' 'Information officer' 'Printmaker' 'Early years teacher'
 'Editorial assistant' 'Advertising account planner' 'Land'
 'Designer, exhibition/display' 'Psychotherapist, dance movement'
 'Oceanographer' 'Geoscientist' 'Homeopath' 'Mechanical engineer'
 'Engineer, communications' 'Soil scientist'
 'Clinical molecular geneticist' 'Arts development officer'
 'Heritage manager' 'Ranger/warden' 'Secretary, company'
 'Designer, industrial/product' 'Academic librarian' 'Comptroller'
 'Psychologist, occupational' 'Television camera operator'
 'Sales executive' 'Psychologist, sport and exercise'
 'Water quality scientist' 'Oncologist' 'Call centre manager'
 'Research scientist (physical sciences)' 'Personnel officer'
 'Clinical embryologist' 'Horticulturist, amenity'

'Geologist, engineering' 'Ambulance person' 'Technical brewer'
 'Health service manager' 'Chief Executive Officer' 'Therapist, art'
 'Librarian, academic' 'Engineer, manufacturing' 'Warehouse manager'
 'Trade union research officer' 'Chartered accountant' 'Media buyer'
 'Surveyor, planning and development'
 'Psychologist, prison and probation services' 'Primary school teacher'
 'Accountant, chartered' 'Engineer, technical sales'
 'Biochemist, clinical' 'Leisure centre manager' 'Accounting technician'
 'Higher education lecturer' 'Science writer' 'Administrator, sports'
 'Planning and development surveyor' 'Psychotherapist' 'Astronomer'
 'Fitness centre manager' 'Quantity surveyor' 'Stage manager'
 'Air cabin crew' 'Theatre director' 'Surveyor, insurance'
 'Commercial/residential surveyor' 'English as a second language teacher'
 'Agricultural engineer' 'Advertising art director' 'Patent attorney'
 'Ceramics designer' 'Sports therapist' 'IT technical support officer'
 'Hotel manager' 'Chartered loss adjuster' 'Radio broadcast assistant'
 'Structural engineer' 'Editor, magazine features' 'Dietitian'
 'Personal assistant' 'Surgeon' 'Administrator, local government'
 'Furniture designer' 'Nurse, learning disability' 'Associate Professor'
 'Physiotherapist' 'Tax adviser' 'Location manager'
 'Commercial horticulturist' 'Conference centre manager'
 'Clothing/textile technologist' 'Designer, fashion/clothing'
 'Public relations account executive' 'Scientist, physiological'
 'Hospital doctor' 'Art gallery manager' 'Product manager'
 'Mental health nurse' 'Engineer, production' 'Human resources officer'
 'Advertising copywriter' 'Insurance account manager' 'Therapist, drama'
 'Microbiologist' 'Immigration officer' 'Bonds trader' 'Tree surgeon'
 'Engineer, electrical' 'Industrial/product designer'
 'Scientist, product/process development' 'Broadcast engineer'
 'Cartographer' 'Computer games developer' 'Research officer, trade union'
 'Development worker, international aid' 'Field trials officer'
 'Transport planner' 'Engineer, water' 'Chief Strategy Officer'
 'Teaching laboratory technician' 'Geologist, wellsite'
 'Air traffic controller' 'Programmer, applications'
 'Commissioning editor' 'Exercise physiologist' 'Embryologist, clinical'
 'Counsellor' 'Retail manager' 'Careers information officer'
 'Production engineer' 'Music tutor' 'Financial adviser'
 'Fisheries officer' 'Audiological scientist' 'Secretary/administrator'
 'Chartered public finance accountant' 'Exhibition designer'
 'Barrister's clerk' 'Actuary' 'Meteorologist' 'Actor' 'Social worker'
 'Programmer, systems' 'Management consultant'
 'Government social research officer' 'Chief Technology Officer'
 'Bookseller' 'Scientist, water quality' 'Quarry manager'
 'Designer, textile' 'Engineer, maintenance' 'Social researcher'
 'Insurance risk surveyor' 'Therapist, horticultural'
 'Nature conservation officer' 'Journalist, broadcasting' 'Paramedic'
 'Petroleum engineer' 'Orthoptist' 'Aid worker'
 'Armed forces operational officer' 'Freight forwarder'
 'Physicist, medical' 'Engineer, chemical' 'Designer, multimedia'
 'Hydrographic surveyor' 'Producer, television/film/video'
 'Scientist, research (maths)' 'Horticultural therapist'
 'Medical sales representative' 'Market researcher' 'Fashion designer'
 'Engineer, building services' 'Pharmacist, community']

 Column: Transaction_Status - Unique Classes: 3
 ['Failed' 'Pending' 'Success']

Column: Fraud_Flag - Unique Classes: 2
 ['No' 'Yes']

Column: Discount_Applied - Unique Classes: 2
[True False]

Column: Transaction_Description - Unique Classes: 5389
['Recently company detail form range a.'
'Anything son baby power heart will not up.'
'Form world around green bar environment pattern.' ...
'Magazine reality share office doctor management outside official.'
'Probably ago western approach.' 'Social suddenly lead.']

```
In [4]: # Check the Fraud Flag classes
if 'Fraud_Flag' in df.columns:
    print("Fraud_Flag Classes:", df['Fraud_Flag'].unique())
    print(df['Fraud_Flag'].value_counts())

# Check Transaction Status classes
if 'Transaction_Status' in df.columns:
    print("Transaction_Status Classes:", df['Transaction_Status'].unique())
    print(df['Transaction_Status'].value_counts())
```

```
Fraud_Flag Classes: ['No' 'Yes']
Fraud_Flag
No      2732
Yes     2657
Name: count, dtype: int64
Transaction_Status Classes: ['Failed' 'Pending' 'Success']
Transaction_Status
Pending    1811
Failed     1789
Success    1789
Name: count, dtype: int64
```

In []: